# Orchestrate or Upgrade? Measure First: A Pre-Registered Cost–Quality Map of Lower-Tier Pipelines vs. Frontier Models Across Five LLM Families

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**Abstract**

Organizations deploying large language models (LLMs) for knowledge work face a budget choice: pay for a frontier model, or orchestrate cheaper models into a multi-step pipeline. This thesis answers that choice with three pre-registered studies that escalate a single question.

**Study 1 — Does structure substitute for tier?** In a 2×2×2 factorial across five LLM families and 19 business problems (1,900 blinded generation–judgment runs), adding a three-step team structure lifts a lower-tier model to **98.8% of frontier quality** at 88% of the cost (T-EGR = 1.49), and structure helps cheaper models significantly more than frontier ones ($d = -0.227$, $p = 0.015$; an effect concentrated in persona-loaded cells and $\approx 0$ without personas). Yet a *single* lower-tier call already captures 98% of frontier quality at just 23% of the cost (T-EGR = 6.15) — the highest-return configuration tested — so the team's added structure buys only a marginal quality gain for about 3.8× the spend. Frontier teams are the worst dollar (3.5× the cost for negligible gain, T-EGR = 0.27). A pre-registered replication on 30 fresh problems and 10 families confirms the core economics out-of-sample; three secondary hypotheses — a marketing-domain exception, a capability-gap correlation, and team-driven variance reduction — failed their pre-registered gates and are not recommended.

**Study 2 — Why substitution stops short.** A refinement pass costing a fraction of a cent per output ($0.001–0.003, family-dependent) recovers up to 64% of the frontier gap but never achieves paired superiority. Qualitative coding ($\kappa = 0.824$) locates the ceiling: orchestration buys *coverage* (more structure, more length) while the frontier wins on *density* (compressed insight, zero boilerplate) — a gap that add-only pipelines cannot close.

**Study 3 — Who should design the pipeline?** When models auto-design their own workflows, authorship barely matters: self-designed and frontier-designed pipelines are statistically indistinguishable in four of five families, and both beat our frozen human-designed instrument. But orchestration economics are family-specific — at 0.24–3.5× solo cost, cheap auto-designed pipelines beat the frontier-solo baseline for Kimi ($d = +0.85$ on novelty, the only family to survive multiplicity control) and lose decisively for Claude ($d = -3.15$). A pre-registered replication on 30 fresh problems finds the pattern transports only at its extremes (Kimi $d = +1.25$; Claude $d = -1.90$), so routing must be *measured per deployment, not looked up.*

**For business decision-makers:** use a *single* lower-tier call by default — the highest return. Add a *pipeline* only where measurement on your own models and problems supports it: effects are family- and problem-specific, and model-authored (auto-designed) pipelines outperform hand-designed ones. Reserve the *frontier* for *density*-critical work — compressed insight with zero boilerplate — the one thing orchestration cannot manufacture. Never combine frontier and team (the worst dollar in the study). Above all, *measure* efficiency (T-EGR) on your own stack rather than trusting any lookup table.

**Keywords:** Large Language Models, Multi-Agent Systems, LLM Orchestration, AI Cost Optimization, Pipeline Architecture, LLM-as-a-Judge, Resource Allocation, Pre-registered Experiment, Cost-Effectiveness Analysis, Business AI Strategy, T-EGR, Model Tier Substitution.

*A practitioner-oriented executive summary with budget-allocation rules appears at the end of the paper (Section A).*

# List of Abbreviations

| Abbr | Definition |
|---|---|
| $BF_{10}$ | Bayes factor in favor of $H_1$ over $H_0$ |
| C-S-N | lower-tier solo, no persona |
| C-S-P | lower-tier solo, persona |
| C-T-N | lower-tier team, no persona |
| C-T-P | lower-tier team, persona |
| F-S-N | frontier solo, no persona |
| F-S-P | frontier solo, persona |
| F-T-N | frontier team, no persona |
| F-T-P | frontier team, persona |
| EGR | Efficiency Gain Ratio |
| FDR | false discovery rate |
| FLOPs | floating-point operations |
| FSN | frontier solo, no persona (baseline) |
| ICER | incremental cost–effectiveness ratio |
| IRR | inter-rater reliability |
| JZS | Jeffreys–Zellner–Siow (Bayes factor prior) |
| LLM | large language model |
| LMM | linear mixed model |
| LOO | leave-one-out |
| OBF | O'Brien–Fleming (spending function) |
| SESOI | smallest effect size of interest |
| T-EGR | Tiered Efficiency Gain Ratio |
| TOST | two one-sided tests (equivalence) |

# 1 Introduction

Recent work has moved from single-model inference to the orchestration of multiple language-model calls: refinement pipelines chain builder, critic, and refiner steps into

structured workflows (Wei et al., 2022; Shinn et al., 2023; Yao et al., 2023); agent teams assign social roles and have models debate toward an answer (Liang et al., 2024); and panels of judges aggregate verdicts across models (Zheng et al., 2023; Verga et al., 2024). Across these designs runs an implicit economic claim — that *structure can substitute for model tier*. A practitioner who cannot afford a frontier model, the argument runs, can reach comparable output quality by composing cheaper models into an orchestrated system. If true, the claim reshapes how organizations price, budget, and architect generative-AI products. If false, orchestration is at best a marginal convenience and at worst a cost multiplier that never reaches frontier quality.

Prior empirical work on generative AI is encouraging but silent on this specific claim. Controlled productivity studies show that a single generative assistant raises output quality and reduces task time (Noy and Zhang, 2023; Brynjolfsson et al., 2023), yet they treat the model as a given, single-agent capability. A field experiment with consultants documented a "jagged technological frontier": large gains inside the model's capability boundary, large losses outside it (Dell'Acqua et al., 2023). Neither study varies the structure around the model. Multi-agent and ensemble work shows that sampling, debate, and tree search can improve accuracy (Wang et al., 2023; Liang et al., 2024; Yao et al., 2023), but the gains are reported without cost-matched comparison to a stronger single model. A nascent theory of test-time compute predicts that, for fixed problem difficulty, more inference compute on a smaller model can outperform less inference compute on a larger one (Snell et al., 2024), but its predictions are in units of FLOPs, not dollars, and only on verifiable tasks.

Four gaps follow. First, no factorial, cost-matched comparison of orchestrated lower-tier models versus frontier solo exists on open-ended tasks. Second, no mechanistic account explains *why* orchestration succeeds or fails. Third, no economic metric is robust to the price-assumption choices that inevitably shape per-run costing. Fourth, orchestration studies almost invariably freeze a *human-designed* pipeline: whether models can design their own workflows, and whether the resulting cost–quality economics transport across model families and fresh problems, is largely untested — leaving practitioners unable to tell whether a winning configuration can be copied or must be re-measured.

This paper addresses all four. We run a pre-registered $2 \times 2 \times 2$ (tier $\times$ structure $\times$ persona) factorial experiment over five model families. The third factor — a fixed cultural persona applied to each pipeline role, carried from our prior work on persona economics (Hariri, 2026b) — is a secondary, boundary question rather than part of the core tier–structure claim; it produces a large negative effect: personas degrade quality sharply with no meaningful cost saving (Section 4.1.2). Nineteen confirmatory business problems are drawn from a 30-problem domain-stratified universe, with 11 problems reserved for feasibility gating per the pre-registered protocol — a separation that keeps feasibility probes from anchoring confirmatory inference (Albers and Lakens, 2018). The experiment produces 1,900 generation–judgment runs, measured by a blinded three-judge panel from families disjoint from every generator. Cost is stored as tokens and re-priced through a frozen price vector under three blended input/output ratios, so every reported dollar is reproducible and re-priceable. A second study applies a cheaper, add-only refinement pass to the orchestrated outputs and codes the qualitative differences between original, refined, and frontier answers, testing the mechanistic prediction that orchestration buys coverage while the frontier wins on density. A third study unfreezes the pipeline itself: across the same five families, models auto-design their own workflows, which we compare against frontier-designed and our frozen human-designed instruments, and we test whether

the resulting cost–quality economics transport to 30 fresh problems in a pre-registered replication.

Six contributions follow:

1. **The first cost-matched, pre-registered factorial** testing whether orchestration structure substitutes for model tier on open-ended business tasks — with negative superiority results reported as primary findings, not buried.
2. **A new economic metric, T-EGR** (quality per dollar, relative to the frontier baseline) and a cost–quality Pareto frontier shown to be robust to price assumptions.
3. **A mechanistic explanation** of why substitution stops short: orchestration buys *coverage* while the frontier wins on *density* — a gap a low-cost refinement pass narrows but never closes.
4. **A pipeline-authorship finding**: who designs the workflow barely matters, but the economics are family-specific, so orchestration must be *measured per deployment, not looked up*.
5. **Practitioner contingency rules** mapping each budget tier and task type to the pipeline architecture with the highest expected return.
6. **External validity — and its boundary**: across two confirmatory replications on fresh problems and a widened family universe, the core cost–quality economics generalize, while the contingent nuances do not, identifying which orchestration findings practitioners can rely on.

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the pre-registered design, apparatus, and analysis, including an AI-assisted research disclosure (Section 3.9). Section 4 presents the three studies as a single buyer's decision journey: Study 1 asks whether structure substitutes for model tier, Study 2 asks why substitution stops short of the frontier, and Study 3 asks who should design the pipeline. Section 5 discusses theoretical and practical implications; Section 6 translates the findings into budget rules and an operating model for practitioners. Section 7 acknowledges limitations. Section 8 outlines future directions. Section 9 concludes.

# 2 Literature Review

This review synthesizes four streams of research: (i) measured productivity effects of generative AI, (ii) multi-agent and ensemble generation methods, (iii) LLM-as-a-judge evaluation methodology, and (iv) open-science methods for economic measurement. We then integrate these streams with our own prior work on personas and efficiency metrics to articulate the research gap that motivates this study.

## 2.1 Productivity effects of generative AI

### 2.1.1 Measured gains in knowledge work

Three controlled studies establish that generative AI raises knowledge-work productivity, but they also reveal that the gains are conditional and bounded. Noy and Zhang (2023) randomized 444 mid-level professionals to ChatGPT-assisted and control arms on professional writing tasks: assistance reduced completion time by 0.8 standard deviations and raised output quality by 0.4 standard deviations, with the largest quality gains among initially lower-performing workers — a compression of the performance distribution. Brynjolfsson et al. (2023) observed 5,179 customer-support agents using a generative

assistant in the field and found a 14% average productivity increase, concentrated among novice and low-skilled workers (+34%), consistent with the model diffusing behaviors of high-ability agents across the workforce.

A critical limitation unites these studies: both treat the model as a *given, single-agent capability*. The unit of analysis is one model call, not an orchestrated system. The professional-writing and customer-support tasks they study are also relatively closed-form: the desired output is well-specified (a support response, a draft document), and quality can be partially verified against ground truth. Open-ended strategic tasks — the domain of this paper — have no verifiable correct answer, which changes both the nature of the "frontier" and the applicability of the productivity gains documented above.

### 2.1.2 The jagged frontier and its boundary

The boundary of these gains is the central caution. Dell'Acqua et al. (2023) ran a field experiment with 758 consultants on realistic professional problems and documented a "jagged technological frontier": inside the model's capability boundary, consultants completed 12.2% more tasks, 25.1% faster, with 40% higher quality; outside it, consultants using the model were 19 percentage points less likely to produce correct solutions than controls. Two productive behavioral strategies emerged ("centaurs" dividing work at the boundary, "cyborgs" interleaving with the model), but neither changed the location of the boundary itself.

The jagged frontier concept is especially consequential for open-ended tasks. In closed-form domains (arithmetic, code, support scripts), the boundary is sharp: a solution is either correct or not. In open-ended business strategy, the boundary is diffuse — there is no single correct answer, only better or worse responses. This diffuseness creates a fundamental ambiguity: when a cheaper model produces a plausible but suboptimal strategy, the user may not know they are "outside the frontier" until the solution fails in deployment. The open question this paper addresses is whether *orchestration structure* can shift the effective capability frontier for these open-ended tasks — and at what cost.

## 2.2 Multi-agent systems, reasoning, and inference-time compute

### 2.2.1 From chain-of-thought to structured reasoning

The modern reasoning literature begins with the observation that language models solve complex problems better when they generate intermediate reasoning steps. Wei et al. (2022) introduced chain-of-thought (CoT) prompting: appending exemplars that show step-by-step reasoning before the final answer elicited comparable reasoning in the model, dramatically improving performance on arithmetic, commonsense, and symbolic reasoning tasks. The CoT paradigm reframed language-model inference from single-shot generation to *structured multi-step deliberation* — a conceptual shift that enabled the orchestration methods reviewed below.

CoT's direct descendant in the accuracy–cost tradeoff space is self-consistency. Wang et al. (2023) showed that sampling a diversity of reasoning paths from a single model and taking a majority vote over final answers improves arithmetic, commonsense, and symbolic reasoning well beyond CoT decoding alone. The result establishes the general principle that parallel sampled calls can be converted into accuracy. Its direct corollary

— that accuracy is bought at a multiplicative cost in calls and tokens — is rarely priced against the alternative of a single stronger model.

### 2.2.2 Interactive deliberation: debate, tree search, and agent loops

Moving beyond independent sampling, recent work has explored *interactive* structures in which model instances influence each other. Liang et al. (2024) showed that multiple LLM instances proposing answers, critiquing each other's responses, and refining their own over debate rounds improves factuality and reasoning on arithmetic and knowledge benchmarks, encourages divergent thinking, and reliably converges to shared answers. Yao et al. (2023) extended the reasoning structure from linear chains to branching trees: Tree of Thoughts treats each reasoning step as a node in a search graph, allowing the model to explore multiple reasoning paths, backtrack from dead ends, and select the most promising branch — deliberate problem solving rather than single-path generation.

These works demonstrate that structure (sampling, aggregation, critique loops, tree search) adds quality beyond a single call. What they do not provide is a cost-matched factorial comparison: prior multi-agent studies report quality gains without measuring dollars, and without comparing orchestrated lower-tier models against an un-orchestrated frontier model on identical problems. The substitution claim — that structure can stand in for model tier — is therefore assumed, not tested.

### 2.2.3 The test-time compute hypothesis

A nascent theoretical framework reframes the model-tier-versus-structure tradeoff as an allocation problem. Snell et al. (2024) demonstrated that scaling inference-time computation (more tokens, verifier-based search, iterative refinement) can be more effective than scaling model parameters for improving performance on reasoning benchmarks. Their compute-optimal curve predicts that for any fixed problem difficulty, there exists an optimal allocation between training compute (model size) and inference compute (test-time search). Below that optimum, a smaller model with more inference compute outperforms a larger model with less; above it, the larger model wins.

This framework is the closest theoretical grounding for the "orchestration economics" claim, but it remains limited in three ways that motivate our study. First, Snell et al. (2024) measure *verifiable* reasoning tasks (math, code) where a verifier can score intermediate steps; open-ended business strategy has no verifier. Second, their test-time compute is *search-based* (sampling + scoring), whereas the multi-agent literature uses *structural* compute (builder–critic–refiner pipelines, debate rounds) — the two may have different cost–quality curves. Third, and most fundamentally, their analysis is in units of FLOPs, not dollars: the practitioner facing an API bill cannot pay in FLOPs. A dollar-matched empirical test is missing.

### 2.2.4 Self-refinement and the limits of iterative improvement

The final reasoning structure relevant to this study is self-refinement: a model critiquing and revising its own output. Shinn et al. (2023) introduced Reflexion, in which a language agent maintains a reflective memory of past failures and uses verbal reinforcement learning to improve subsequent trials. On coding and decision-making tasks, Reflexion outperformed single-shot baselines and even standard CoT by learning from its own errors.

Self-refinement raises a hypothesis directly tested in our Study 2: if a cheaper model's initial output is below the frontier, can a low-cost refinement pass close the gap? The multi-agent and self-refinement literatures imply that iterative improvement should raise quality. But they do not tell us whether the improvement is sufficient to reach parity with a frontier model that was not itself refined — the marginal comparison that determines substitution. Nor do they tell us *why* refinement succeeds or fails on open-ended tasks. Our qualitative coding protocol (Section 3.6.5) is designed to answer this mechanistic question.

## 2.3 LLM-as-a-judge evaluation methodology

Because our outcomes are open-ended business solutions, evaluation itself is a design object. Zheng et al. (2023) established LLM-as-a-judge as a scalable evaluation paradigm, showing strong agreement between strong judge models and human preferences on pairwise comparison tasks, while documenting three systematic biases that threaten validity: position bias (preferring the candidate presented first), verbosity bias (preferring longer outputs), and self-enhancement bias (preferring outputs from the same model family as the judge). Their mitigations — swapping candidate order, controlled few-shot examples, length normalization — inform our blinding and order-shuffling protocol.

The self-enhancement bias is particularly acute for multi-agent studies. If the generator and judge share model architecture, the judge may reward the generator's stylistic conventions rather than substantive quality. Verga et al. (2024) extended the single-judge design into a jury: a Panel of LLM evaluators (PoLL) composed of several smaller models from different families correlated better with human judgments than a single large judge, at lower cost, while mitigating the self-preference bias that afflicts same-family evaluation. Our evaluation framework inherits this logic directly: three judges from families disjoint from all generators, each under three frozen prompt variants, scoring blind to cell and family. Because no judge shares a family with any generator, the "persona penalty" reported in Section 4.1.2 cannot be attributed to same-family self-enhancement bias (Zheng et al., 2023).

## 2.4 Open-science methods and economic measurement

The fourth stream supplies the discipline this paper applies. Nosek et al. (2018) argue that preregistration distinguishes confirmatory from exploratory analysis, converting transparency into credibility; our protocols, including the four modification protocols (05–08), were frozen before the corresponding computations ran and are timestamped in the OSF registration chain (Table 2), which also discloses the two exceptions. Steegen et al. (2016) show that conclusions can vary across reasonable analytic choices and recommend reporting a multiverse of analyses; our price-assumption panel (three blended input/output ratios) and pre-registered robustness battery implement this idea. Sandve et al. (2013) codify reproducible computational research — versioned scripts, recorded steps, stable environments — which our token ledger and frozen configuration files follow.

Two statistical practices complete the foundation. Albers and Lakens (2018) demonstrate that effect sizes estimated from pilots are unstable and bias follow-up power analyses, so pilot data must never anchor confirmatory claims; accordingly, our pilot data informed only feasibility gating, never the confirmatory analysis. Lakens (2013) makes the case for reporting effect sizes with confidence intervals to support cumulative science;

we report Welch's $d$ with bootstrap intervals throughout. Cohen (1960)'s $\kappa$ coefficient provides our inter-rater and coder-agreement metric.

Finally, measuring the economics of inference requires separating quantity from price. Diewert (1976) formalized the decomposition of value changes into price and quantity components in index-number theory; we apply the same logic by storing tokens (quantity) and multiplying by a frozen price vector, so every reported dollar is reproducible and re-priceable. Tomlinson et al. (2024) quantify the environmental footprint of generative tasks, reinforcing that inference cost — economic and environmental — is a first-order design variable rather than an implementation detail.

## 2.5 Integration: from persona economics to orchestration economics

Our own prior work supplies the immediate empirical background. In a cross-architecture evaluation of cultural personas across seven large language models, persona effectiveness followed a contingency model rather than a universal rule, and the Efficiency Gain Ratio (EGR) showed that small models with personas achieved superior performance at 40–88% lower cost (Hariri, 2026b). The same line of work measured creativity with multi-judge panels over thousands of generated business solutions, establishing Novelty as the best-agreed creativity dimension (Hariri, 2025a), and showed that team composition effects are contingent on task verifiability (Hariri, 2025b).

Integrating the four streams above with this lineage yields the gap and a pre-registered hypothesis set spanning all three studies, each element grounded in the literature or in our prior work. The Study 1 labels $H_1$–$H_5$ follow the registered protocol exactly (OSF registration `osf.io/mjb8x`, Protocol 01): $H_1$ is the primary tier × structure interaction; $H_2$, $H_3$, and $H_4$ are the tier, structure, and persona main effects; $H_5$ is the structure × persona interaction. Studies 2 and 3 add their own pre-registered hypotheses below (Protocols 09 and 13).

**Substitution hypothesis ($H_1$):** The test-time compute framework (Snell et al., 2024) motivates the prediction that structure can substitute for model tier at the lower-tier end of the compute-optimal curve. We test this prediction with a factorial tier × structure design, priced in dollars, on open-ended tasks where no verifier exists. If the hypothesis holds, team structure should raise quality more for cheaper models than for frontier models.

**Researcher-intent note.** As the registered rationale itself states (Protocol 01), the motivating comparison is direct: *cheap team+persona versus a frontier model running solo*. The direct comparison is pre-registered as the economic analysis (Protocol 04, every cell priced against the F-S-N baseline) and reported in Section 4.1.3.

**Main-effect hypotheses ($H_2$, $H_3$):** The registered protocol also specifies the tier main effect ($H_2$: frontier models score higher than their lower-tier siblings, averaged over structure and persona) and the structure main effect ($H_3$: teams score higher than solos, averaged over tier and persona). We report both as secondary context for the economic analysis.

**Persona hypotheses ($H_4$, $H_5$):** Our prior work (Hariri, 2026b) showed that cultural personas shift quality within single-model cells. $H_4$ extends the registered claim to this design: persona-loaded conditions should score higher than no-persona conditions, averaged over tier and structure. $H_5$ adds that the persona effect should be larger for team

structures than for solo structures (personas amplify structural rigor). If personas primarily add stylistic variation without substance, the persona-loaded cells should instead be Pareto-dominated by their no-persona twins.

**Refinement hypothesis (Study 2, Protocol 09):** The self-refinement literature (Shinn et al., 2023) suggests that a low-cost critique-and-revise pass can recover quality gaps. Study 2 tests whether a frozen, add-only refiner can close the residual gap between orchestrated lower-tier outputs and frontier outputs — and whether the mechanism is additive coverage (more structure) or compressed insight (better density). Study 2 was pre-registered as a separate study before any refinement run, with its own feasibility gate (G1), superiority gate (G2), and three directional criteria (H2.1–H2.3) frozen in the decision log (Protocol 09, decisions D39–D40).

**Authorship hypotheses (Study 3, Protocol 13):** Every pipeline in the literature above — and in Studies 1–2 — is *human-designed*; the registered-but-unexecuted A7 arm isolated authorship as the one design variable left untested, and Study 3 executes it. Across the same five families, the cheap model either authors and runs its own workflow (D-SELF) or runs one authored by the frontier flagship (D-FRONT). The primary hypothesis is *directional*: frontier-authored pipelines outperform self-authored ones (H-D1: D-FRONT > D-SELF), on the ground that design capability transfers through authored structure — the same logic by which a cross-family refiner beat within-family editors in the Study 2 extension. Two further hypotheses complete the set: that model-authored pipelines are at least as good as the frozen human-designed instrument (H-D2, two-sided), and that frontier-authored design stays cheaper per quality point than frontier solo once design tokens are charged (H-D3, economic). A pre-registered confirmatory replication (Protocol 16) then converts the family-level effects into binding directional gates, testing whether the cost–quality ordering transports to fresh problems.

Section 3 specifies the pre-registered design, apparatus, and analysis plan that test this hypothesis set, with every run priced in dollars; Section 4 reports the outcomes. One hypothesis is mechanistic rather than purely economic: the qualitative coding protocol (Section 3.6.5) tests the refinement hypothesis's prediction that orchestration buys *coverage* while the frontier wins on *density*, by coding original, refined, and frontier answers against a frozen codebook.

# 3 Methodology

## 3.1 Research Design

The experiment is a $2 \times 2 \times 2$ between-cell factorial with model *family* as a blocking factor. The three manipulated factors are **tier** (frontier flagship vs. lower-tier sibling within the same generation), **structure** (solo single call vs. three-step team of builder → critic → refiner), and **persona** (none vs. a fixed cultural-persona mapping applied to each team role). Five model families — Kimi, Qwen, GPT-5.6, Claude, and Gemini — supply both frontier and cheaper tiers. Thirty business problems, stratified across knowledge-work domains, are identical across all cells; no instruction wrappers are added to the rendered prompt. Every design and statistical decision is time-stamped, numbered (D1–D60), and registered in a frozen decision log; sixteen protocols — twelve standalone and four modifications of the original registration (05–08) — were registered before the corresponding

computation ran, maintaining the confirmatory/exploratory separation required by pre-registration discipline (Nosek et al., 2018).

Two terms recur throughout the paper and are worth fixing precisely. A **protocol** is a single pre-registered experiment plan: it states the research question, the hypotheses, the models and problems to be run, the analysis code, and the stopping rules — and it is frozen (time-stamped and registered on OSF) *before* any data for that question are generated, so that results cannot be retrofitted to expectations. An **arm** is one experimental condition inside a protocol — one specific combination of model tier, pipeline structure, and persona setting that produces a batch of runs (e.g. "lower-tier team without persona" is one arm). Section B lists every protocol and arm with its role and status.

This paper reports three studies and two pre-registered replications. **Study 1** is the factorial experiment above ($H_1$–$H_5$; Protocols 01–04 and their modification Protocols 05–08). **Study 2** asks whether a cheap *refinement pass* on an existing answer can close the remaining gap to frontier quality (Protocol 09), followed by qualitative coding of what orchestrated and frontier outputs actually contain (Protocol 10). A pre-registered **Study 2 extension** (Protocol 11) repeats the refinement test with two further families (GPT, Gemini) and adds a refiner-identity contrast: each family refined by its own model versus a *cross-family* refiner editing another family's output. The Study-1→Study-2 sequence follows an explanatory-sequential loop (D39): Study 2's *structure* (universal add-only refiner prompt, paired design, power rule) was frozen on 2026-07-30 — before Study 1's main run — while its *repair targets* (the dimensions refinement must repair or protect) were set from Study-1 results and locked before any refinement run. **Study 3** manipulates pipeline design authorship with model-authored pipelines (Protocol 13, Section 4.3). The two replications stress-test the core claims out-of-sample: a confirmatory wave on 30 fresh problems across ten families (Protocol 12, OSF jg49x; Section 4.1.7), and a transport test of Study 3's family-level effects on new problems (Protocol 16; Section 4.3.1).

Table 2: OSF registration chain (Open-Ended Registrations branched from project node `8pwej`). The text cites Protocol numbers; this table is the authoritative GUID map.

| Study / Protocol | OSF | Registered | Role |
|---|---|---|---|
| Study 1 (Protocols 01–04, A1/A2) | mjb8x | 2026-08-06 | factorial pre-registration[†] |
| Study 2 extension (Protocol 11) | 83c9u | 2026-08-22 | refinement extension (GPT, Gemini) |
| Wave (Protocol 12) | jg49x | 2026-08-23 | Study 1 confirmatory replication |
| Study 3 (Protocol 13, v2) | bwc93 | 2026-08-29 | authorship study (operative)[‡] |
| Study 3 (Protocol 13, G1) | 639eq | 2026-08-29 | feasibility-gate clarification |
| Study 3 replication (Protocol 16) | zkm8v | 2026-08-30 | family-moderation transport test |

[†] mjb8x *memorializes* the Study 1 pre-registration: the protocol and decision log were frozen before data collection, but this OSF timestamp post-dates main-run generation (as the registration itself discloses).
[‡] bwc93 is the operative Study 3 registration, superseding the v1 registration (9ukjz) after a pre-analysis instrument audit; the 73 v1 runs were archived and never pooled.

**Registered but unexecuted arms (disclosure).** The Study-1 pre-registration also specified four special mechanism arms — A5 (a critic-gated WHILE loop on the cheap team), A6 (a cost-matched frontier-solo baseline), A7 (an adaptive per-problem pipeline designer), and A8 (a shuffled persona-to-role mapping) — each with a pre-registered feasibility/futility gate. These gates were never evaluated and the arms were never executed within Study 1: the main run was scoped to the eight cells of the factorial design itself (2 tiers × 2 structures × 2 persona settings, listed in Section B), and the Study-2 exten-

sion and the replication wave took budgetary priority. Per no-file-drawer discipline we disclose their registered-but-unexecuted status here rather than silently dropping them. The design-authorship question behind A7 has since been executed as Study 3 (Protocol 13; Section 4.3); the loop question behind A5 is the subject of a separate pre-registered protocol (Section 8).

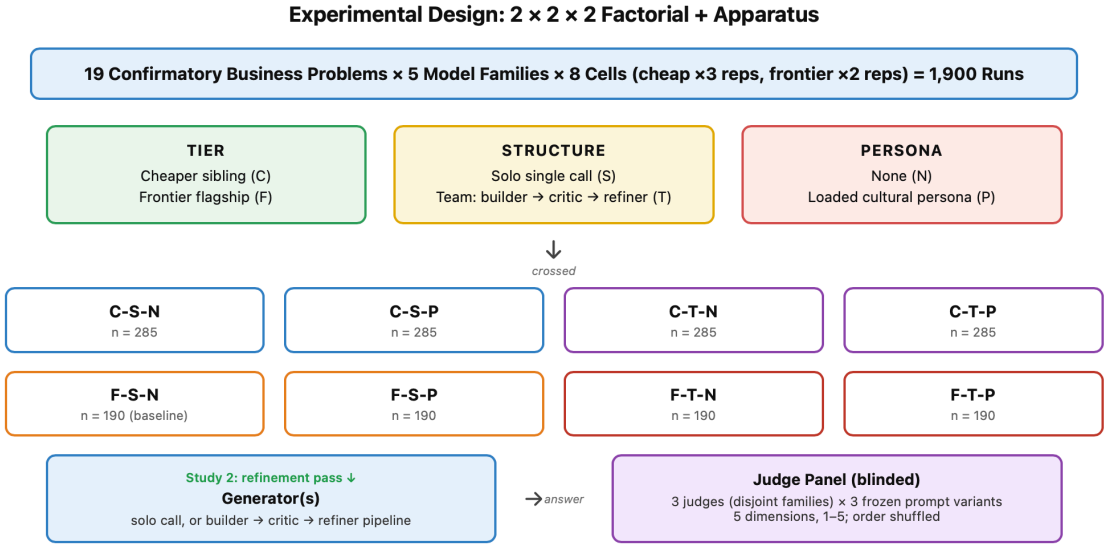



Figure 1: Experimental Design: $2 \times 2 \times 2$ Factorial + Apparatus.

## 3.2 Models Evaluated

### 3.2.1 Frontier Models (Five Families)

Each family contributes its current *mainline* flagship, pinned on 2026-07-30: `kimi-k3`, `qwen3.7-max-2026-06-08`, `gpt-5.6-sol`, `claude-opus-5`, and `gemini-3.1-pro-preview`. Above-mainline tiers (e.g. `claude-fable-5`) are deliberately excluded because they would break cross-family comparability. Gemini's preview status carries a pre-registered fallback to `gemini-2.5-pro` if the model is retired mid-experiment; the fallback was never invoked. All models run with generator temperature = 1, a uniform value forced by the OpenAI and Moonshot APIs (which reject other values) and applied identically across families for *ceteris paribus* comparability (D27).

### 3.2.2 Lower-Tier Siblings

Each family contributes one lower-tier model from the same generation: `kimi-k2.6`, `qwen3.7-plus-2026-05-26`, `gpt-5.6-luna`, `claude-haiku-4-5-20251001`, and `gemini-3.1-flash-lite`. Price ratios (frontier : cheaper) range from 3.1× to 8× across families. The size of this price gap differs from family to family; our statistical model simply gives each family its own baseline (a *family random effect*, Section 3.6.2), so these differences in price gap cannot distort the main tier × structure comparison.

### 3.2.3 Replication-Wave Models (Five Additional Families)

The confirmatory wave (Protocol 12, Section 4.1.7) repeats the three core cells (C-S-N, C-T-N, F-S-N; persona excluded by design, D50) on 30 fresh problems and adds five families beyond the core set, pinned on 2026-08-22/23 per Protocol 12 §3: Thinking Machines (`Inkling` / `Inkling-Small`), OpenAI open-weight (`gpt-oss-120b` / `gpt-oss-20b`), Google open-weight (`gemma-4-31B-it` / `gemma-3n-E4B-it`), Mistral (`mistral-medium-2604` / `mistral-small-2603`), and MiniMax (`MiniMax-M3` / `MiniMax-M2.7`). The five core families run their pinned Study-1 models unchanged, with one disclosed routing-only deviation: the Kimi lane runs on the native Moonshot provider because the Together-hosted `kimi-k2.6` was retired before the wave (D48; identical prompts and decoding, provider route auditable in the ledger).

## 3.3 Orchestration Structures and Personas

**Solo** runs invoke a single model call on the rendered problem. **Team** runs execute a three-step pipeline — builder produces a draft, critic critiques it, refiner produces the final answer — with all intermediate outputs discarded from the judged answer (and therefore from the quality score), while the cost ledger records every step: the per-run token total is the sum of the builder, critic, and refiner calls, so team costs include the full pipeline, not just the final step. The one gap is retry accounting: when a step failed and was retried, the failed attempt itself logged zero tokens and zero latency (an instant transport rejection, consuming nothing billable), but on 110 of the 950 completed team runs the ledger's run total then omitted tokens from *completed* earlier steps, so recorded cost understates true cost on those runs; we bound this in Section 7. Persona is implemented as a fixed mapping from role (builder/critic/refiner) to a cultural persona description, identical across all problems and families (Hariri, 2026b). The mapping is pre-registered and never varied per-problem; it is therefore a treatment level, not a confound.

## 3.4 Stimuli and Response Generation

Thirty domain-stratified business problems are used. Fifteen are inherited verbatim from a prior validated set (Hariri, 2025a); the remaining fifteen were drafted for this study so that the full grid covers six domains with five problems apiece (the post-cutoff contamination-control cases belong to the replication wave's fresh problem set, Section 4.1.7). Each problem is rendered as `background_context + "\n\n" + business_problem` with no instructional wrapper — adding wrapper text would introduce an unmeasured prompt treatment. Generator temperature is fixed at 1 (Section 3.2). Every run is time-stamped and flagged for whether it executed inside the pre-registered preferred window (14:00–00:00 UTC), which captures the Qwen off-peak discount; window-free providers (Kimi, Together-hosted models, Western providers) are time-invariant and therefore unaffected. DeepSeek pricing was stable throughout the experiment (no peak-surge window was in effect during the execution period).

## 3.5 Evaluation Framework

### 3.5.1 Judge Model Selection

Three judges from families disjoint from every generator: `deepseek-chat` (resolved to `deepseek-v4-flash`), `zai-org/GLM-5.2`, and `meta-llama/Llama-3.3-70B-Instruct-Turbo`. GLM-5.2 is a within-family substitution for the preregistered GLM-4.7 (non-serverless on Together at pilot day, D33); judge prices are recorded in the frozen price vector for completeness only — judge cost is excluded from all cell economics. The disjointness rule (D33) guards against self-enhancement bias (Zheng et al., 2023): even when deepseek is the registered fallback generator, the fallback was never invoked, so disjointness holds in the executed data.

### 3.5.2 Scoring Rubric

Five 1–5 dimensions are scored: **Novelty, Usefulness, Flexibility, Elaboration, and Cultural Sensitivity**. The composite primary outcome is the mean of the five dimensions. Three prompt variants are used per judge, frozen verbatim and never re-optimized after a prior judge-prompt optimization attempt collapsed $\kappa$ in earlier work (Hariri, 2025a). Every solution is scored by all three judges under all three variants (nine scores per solution), with candidate order shuffled per variant to neutralize position bias (Zheng et al., 2023). Judges run at temperature 0 for determinism.

### 3.5.3 Inter-Rater Reliability

Pairwise judge correlations on per-run composite scores are 0.87 (deepseek–zhipu), 0.61 (deepseek–meta), and 0.49 (meta–zhipu). The panel-level reliability exceeds any single judge's correlation with the latent quality; this mirrors the jury logic of Verga et al. (2024) in which diversity of smaller judges outperforms a single large judge.

## 3.6 Statistical Analysis

### 3.6.1 Effect Sizes

Welch's $d$ with bootstrap 95% confidence intervals (5,000 resamples, seed 42) throughout (Lakens, 2013). Estimation-first reporting: intervals, not $p$-values, carry the weight of the argument; $p$-values are reported for completeness and to comply with sequential-analysis boundaries (Section 3.6.2).

### 3.6.2 Inferential Statistics

The primary test of $H_1$ is the pre-registered interaction contrast $d = (FT - FS) - (CT - CS)$, expected negative (team structure helps cheaper models more than frontier models). Inference uses a mixed model with random effects for *problem* and *family*, solving the pseudo-replication identified in the author's prior work (Hariri, 2026b). Sequential analysis with an O'Brien–Fleming spending function (protocol 03) governs interim looks and the final stopping boundary; the Bayesian JZS Bayes factor is reported as recommended and pre-registered in the robustness battery[1] but does not gate the decision. Family-wise

[1] Protocol 06 — Extended Interim-2 Analysis Methods, items 6.1–6.7; OSF registration `https://osf.io/mjb8x`.

corrections within secondary hypotheses use Benjamini–Hochberg FDR ($\alpha = 0.05$); Holm is used for the small set of directional family comparisons in Study 2.

Later studies and replications follow the same estimation-first discipline with study-specific designs. The **Study 2 extension** (Protocol 11) uses paired within-problem tests with Holm adjustment across arms, plus a two-sided refiner-identity contrast (no pre-committed direction). The **wave** (Protocol 12) fits an LMM with random intercepts for family and variance components for problem and family × problem, evaluated against binding gates W1–W7; when the dual-variance-component model was numerically singular, a pre-declared fallback substituted a parsimonious structure with bit-identical fixed effects (D52). **Study 3** (Protocol 13) tests the authorship contrast with paired $t$-tests within family × problem, Holm-adjusted across the five families, Novelty as the primary dimension and the composite as a robustness variant (conclusions are DV-invariant). **Protocol 16** converts Study 3's point estimates into binding directional gates: problem-level Welch tests of D-SELF against the frozen wave F-S-N baseline, one-sided $p$ by normal approximation, and a TOST equivalence gate for GPT-5.6 (90% CI within ±0.30), executed by a hash-locked analysis script frozen before judging.

### 3.6.3 Economic Analysis

Cost is stored as tokens in a run-level ledger and re-priced through a frozen price vector (`config/price_vector.json`, version 2026-07-30). Because the ledger does not split input from output tokens, cost is computed under a blended price $\text{blend}(\varphi) = \varphi \cdot p_{\text{out}} + (1-\varphi) \cdot p_{\text{in}}$ with a primary $\varphi = 0.50$ and a sensitivity panel $\varphi \in \{0.25, 0.50, 0.75\}$ (Diewert 1976 provides the price–quantity decomposition precedent; Steegen et al. 2016 the multiverse-robustness precedent). Frontier-membership claims must be stable across all three $\varphi$. Studies 3 and the P16 confirmatory use a newer ledger that records prompt and completion tokens separately for every call, so their costs are priced exactly at provider list rates; the blended-$\varphi$ treatment applies to Studies 1–2 and the wave, where only total tokens were logged. The **Tiered Efficiency Gain Ratio (T-EGR)** is $(\text{Quality}_{\text{cell}}/\text{Quality}_{F\text{-}S\text{-}N})/(\text{Cost}_{\text{cell}}/\text{Cost}_{F\text{-}S\text{-}N})$, computed within each family and pooled; F-S-N is the baseline. **ICER** is reported relative to the lowest-cost cell. Judge cost is excluded from unit T-EGR (it is shared infrastructure, not cell compute; a disclosed exclusion).

### 3.6.4 Robustness Checks

The pre-registered robustness battery for Study 1 (Protocol 06, items 6.1–6.7) comprises bootstrap confidence intervals on all primary contrasts (6.1), Benjamini–Hochberg FDR on secondary contrast families (6.2), a Type-II factorial ANOVA sensitivity layer with partial $\eta^2$ (6.3), JZS Bayes factors for non-significant contrasts (6.4), leave-one-family-out sensitivity (6.5), leave-one-problem-out generalizability (6.6), and a sensitivity power analysis in place of post-hoc power (6.7); all seven are reported in Section 4.1.5. After the data lock, an explicitly exploratory modification (Protocol 07, registered in the decision log but not part of the OSF package) added five analyses (7.1–7.5), all reported with the results: the random-effects meta-analysis across the five Study-1 families (7.4, Section 4.1.5), the persona dimension decomposition (7.5, Section 4.1.2), and a mixed-model refit, a TOST equivalence test, and empirical-Bayes partial pooling (7.1–7.3, Section 4.1.5). All Protocol-07 results are labeled exploratory wherever reported. Study 2 uses paired within-family gates with a novelty-protection non-inferiority margin and a

cross-family refiner check (Section 4.2); the wave applies the six pre-registered W-gates (Section 4.1.7); Study 3 carries the G1 feasibility and G2 superiority-stop gates plus a composite-DV robustness check (Section 4.3); and the P16 confirmatory applies five binding replication gates (Section 4.3.1).

### 3.6.5 Qualitative Coding Protocol (Study 2)

Thirty triples (original CTP / refined CTP / same-problem same-family FSN) are sampled deterministically (seed 20260801, 10 per family). Two blinded LLM coders perform a two-wave coding: free induction on a subset, then closed coding against a frozen 13-code codebook.[2] The codebook is organized around three research questions: **RQ1** asks whether orchestration adds *insight* (compressed reasoning, named concepts, diagnostic reframing) — six codes; **RQ2** asks whether orchestration adds *structure* (scaffolding, timelines, matrices) — six codes; **RQ3** asks whether orchestration adds *bloat* (redundancy) — one code. Cohen's $\kappa$ gate is 0.70; codes below threshold are downgraded to medium reliability and disclosed (Cohen, 1960). The prompt presented to coders contains the three texts only — no cell label, family, or score, preserving blind evaluation.

## 3.7 Validity and Reliability

**Construct validity:** the rubric separates Novelty from Usefulness/Elaboration, so a stylistic shift (e.g. more discourse markers from a persona) is not conflated with substantive quality. **Internal validity:** frozen configurations, blinded judging, shuffled order, disjoint judge families. **External validity:** single task domain (English business strategy) is the acknowledged boundary. **Disclosed threats.** (i) *Retry cost accounting.* 234 of the 950 completed team runs contain a failed-step record; in 124 of these the failed attempt logged zero tokens and the recorded total is complete. On the remaining 110 runs, a retried step caused the ledger's run total to omit tokens from completed earlier steps, so reported team costs understate true cost on those runs (median true-to-recorded ratio 1.69; 4.4% of team-cell cost in aggregate). The bias favors the team cells and therefore favors the paper's central claim. It is disclosed but not corrected, since correcting it could only shrink, not reverse, the cost advantage. (ii) *Truncation at source.* A small set of very long runs was capped by the transport layer before storage; each run records whether this happened, and the affected runs are audited separately rather than silently excluded.

## 3.8 Code and Data Availability

The complete replication bundle is public at https://github.com/youssef-hariri/orchestration-economics: all analysis code; the frozen token ledgers; judge prompts and rationales; the decision log (D1–D60); Protocols 01–16; the problem banks; and the run-level responses for every study (main run, 1,900 runs; replication wave, 2,418; Study 3, 544; Protocol 16, 441). Every prompt used in any run is extracted verbatim into `config/prompts/` (13 files, with machine-checked provenance): the Study-1 role templates and neutral solo prompt, the Study-2 add-only refiner prompt, the Study-3 platform-verbatim designer and repair prompts, the three judge variants, and the three personas. Study-3 auto-designed pipelines are published as per-run JSON (`runs/study3/designs/`),

[2] Codebook development followed the pre-registered qualitative analysis protocol (OSF registration https://osf.io/mjb8x, file `02-QUALITATIVE-ANALYSIS-PROTOCOL.md`).

each listing every agent's name, role, and full system prompt. Team intermediate outputs (builder/critic drafts) were not persisted — only final answers were stored and judged; this disclosed design choice bounds what re-analysis can recover (Section 7).

The repository additionally ships a machine-readable **claim-to-evidence map** (`paper/claim_map/`) 32,992 datapoints extracted from the frozen results files, each with source file, key path, and verbatim excerpt, plus 161 paper mentions with verbatim quotes linked to their datapoints. Running `verify_two_way.py` re-checks every datapoint against the raw files (exit code 0 at submission). The pre-registration trail is the OSF registration chain mapped in Table 2: each OSF registration archives its protocol file, and the table discloses the two exceptions — the Study 1 memorialization (mjb8x) and Study 2 (Protocols 09–10), which are committed to the repository and frozen in the decision log (D39) but were not separately OSF-registered. A materialized results layer (D54) ships every statistical test as a flat CSV — per-solution judge scores for all studies, the wave gate statistics, the heterogeneity batteries, and the qualitative codebook with per-code $\kappa$ — so that no number in this paper requires re-running code to inspect. A reproduction runbook (Sandve et al., 2013) lists the commands that regenerate every number reported in Section 4.

## 3.9 AI-Assisted Research Disclosure

This study used large language models as research tools under continuous human supervision. Specifically, **Kimi** (Moonshot AI, `kimi-2.6 and kimi-k3`) and **Qwen** (Alibaba Cloud, `qwen3.7-max and qwen3.8-max`) were employed in the following capacities: (1) **ideation** — brainstorming experimental design alternatives, hypothesis formulations, and analysis strategies during the protocol-drafting phase; (2) **code generation** — producing initial versions of the execution pipeline, statistical analysis scripts, and figure-generation code, all of which were reviewed, corrected, and approved by the author before execution; (3) **audit** — reviewing citation accuracy, checking statistical assumptions, and identifying inconsistencies across protocol documents; (4) **execution** — running the generation and judging campaigns through API calls, with all configurations frozen and logged before each run. Higher-tier models were employed for ideation, audits, and task execution; lower-tier models were employed for code generation.

No LLM made unsupervised decisions about the experimental design, statistical analysis, or the interpretation of results. All sixteen pre-registered protocols, all 60 design decisions (D1–D60), and all analysis choices were authored, reviewed, and approved by the human author. Outputs were validated against expected ranges, and every reported number traces to a frozen, version-controlled analysis pipeline. The qualitative coding protocol (Section 3.6.5) used two LLM coders with blinded evaluation — a design choice that converts the LLM's systematic biases into a measured inter-rater reliability coefficient ($\kappa = 0.824$) rather than an unmeasured confound.

A concrete oversight episode illustrates why continuous human audit is not procedural decoration. When the AI agent implemented the Study 3 design-authorship instrument, it silently introduced two deviations from the product feature under study (DeeWaanAI's auto-designer): it hard-capped pipelines at four steps (the platform allows up to seven) and wrote a bespoke designer prompt, whereas the platform uses a fixed internal prompt with free-form, task-adapted roles. Both deviations were invisible in the code's behaviour — every pilot design complied, so no error surfaced — and were caught only when the human author audited the instrument's rationale against the platform source code before confirmatory data collection. The instrument was corrected and the correction registered

*before* the Study 3 campaign (D56). The lesson for researchers delegating to AI agents: an AI assistant's silent defaults are themselves experimental-design decisions, and they diverge from the intended construct precisely where the researcher did not specify — audit the rationale behind every parameter, not only the parameters named in the protocol.

A second, subtler episode concerns operationalization rather than intent. The registered rationale states the author's motivating comparison directly (cheap team+persona versus frontier solo; Section 2.5); it was the AI assistant that chose to test it as a pooled tier × structure interaction across all eight factorial cells — a valid test, but one whose machinery pools cells the motivating question never asks about. The intent was the author's and is on the record; the choice of test statistic was the assistant's. The lesson generalizes: audit how an AI assistant operationalizes a research question — hypotheses, prompts, parameters — not only the outputs it produces.

A third episode concerns the paper itself. During preparation of the machine-readable claim-to-evidence map that accompanies this paper (every quantitative claim linked to its frozen source file), one claim — the per-family CTP-vs-F-S-N paired contrasts reported in an earlier draft ($d = -0.71, -1.42, -2.17$ for kimi, claude, qwen) — could not be traced to any frozen results file or analysis script; it survived from an early exploratory computation whose composite convention had since changed. Recomputed from the frozen ledger with the study's fixed conventions, the values are materially different for two of five families (Section 4.1.3), and the paper was corrected. The lesson: every number printed in a manuscript must be traceable to the experiments' raw, frozen data — not to an analyst's (or an AI assistant's) memory of an earlier run. Claim-to-evidence tracing, human-audited, is the only reliable way we found to enforce this.

The use of LLMs as evaluation judges (Section 3.5) is a distinct methodological choice from the use of LLMs as research tools: the judges evaluate *generated business solutions*, not the research process itself. The disjointness rule (D33) ensures that no judge shares architecture with any generator, mitigating the self-enhancement bias documented by Zheng et al. (2023).

**Platform use and conflict-of-interest disclosure.** All pipeline generation and execution in Studies 1–3 ran on **DeeWaanAI** (`https://deewaanai.com`, accessed August–September 2026), a commercial orchestration platform; the auto-designer evaluated in Study 3 and Protocol 16 is DeeWaanAI's production feature, used unmodified (fixed internal design prompt, free-form roles, up to seven steps with critic-gated loops). The author is affiliated with DeeWaanAI, and the findings directly inform the platform's routing and persona features (Section 9). This relationship is disclosed so readers can weight the translational claims accordingly; all protocols, gate outcomes (including failed ones), and attrition events are pre-registered and reported as-is, and the full data and code — including all pipeline definitions, prompts, raw model outputs, judge calls, and analysis scripts, sufficient to reproduce every reported statistic without access to the platform — are publicly available at `https://github.com/youssef-hariri/orchestration-economics`.

# 4 Results

This section reports three studies that escalate a single business question — *orchestrate cheaper models, or pay for the frontier?* **Study 1** (Section 4.1) tests whether pipeline structure substitutes for model tier and prices the answer in dollars. **Study 2** (Section 4.2) explains *why* that substitution stops short of the frontier — its refinement stage was frozen before Study 1's main run (Protocol 09, D39), while its repair targets were set

from Study-1 results and locked before any refinement run, an explanatory-sequential loop disclosed in Section 3 (D39). **Study 3** (Section 4.3) asks whether it matters *who designs* the pipeline. Each study's headline claims are then stress-tested out-of-sample in a pre-registered replication, so the rules we recommend are the ones that survived contact with new data — and the ones that did not are reported as failures, not buried.

## 4.1 Study 1: Does Structure Substitute for Model Tier?

### 4.1.1 The interaction effect: significant as registered, but persona-conditional

The pre-registered statistical test of the substitution thesis is the tier × structure interaction: does team structure lift cheaper models more than frontier ones? The pre-registered interaction contrast is $d = -0.227$ (bootstrap 95% CI $[-0.410, -0.040]$, Welch $t = -2.45$, $p = 0.0145$), with the sequential-analysis boundary crossed at the scheduled final look. A pre-registered secondary decomposition (Protocol-06 item 6.2) qualifies where this effect lives: within no-persona cells the interaction is essentially zero ($d = -0.006$, n.s.), while within persona cells it is large ($d = -0.390$, $p_{\text{FDR}} = 0.007$; tier × structure × persona three-way $-0.151$, $p < 0.001$, exploratory Protocol-07 item 7.1). Under personas every cell is depressed, and inside that lowered field teams rescue cheap models (+0.117) while slightly dragging frontier ones (−0.036); without personas, teams lift both tiers by the same small amount (+0.02). For a firm making an AI budget decision, the decision-relevant question is therefore the direct comparison — how close a cheaper team gets to frontier solo, and at what price — answered in Section 4.1.3 and Section 4.1.4; in the default no-persona configuration, the case for cheap teams rests on their cost advantage, not on amplified returns. The JZS Bayes factor for the interaction is 1.002 (inconclusive), reported as recommended but not gating the decision.

The secondary effects provide additional strategic context. The **tier main effect** ($H_2$: $d = +0.294$, CI $[+0.201, +0.382]$, $BF_{10} = 1.7 \times 10^7$) confirms that frontier models command a quality premium, but the magnitude is dwarfed by the cost difference. The **structure main effect** ($H_3$: $d = +0.117$, CI $[+0.030, +0.205]$) shows that teams slightly outperform solos on average — a small but consistent productivity gain from orchestration. The **persona main effect** ($H_4$) is large and in the *opposite* direction to the pre-registered expectation: $d = -0.691$ (CI $[-0.761, -0.620]$, $BF_{10} = 1.0 \times 10^{50}$). Persona-loaded cells score *lower* than their no-persona twins, representing a failed investment in stylistic differentiation. From a business perspective, personas degrade output quality sharply while reducing cost only marginally (persona cells emit fewer tokens) — a value-destroying configuration. An exploratory dimension decomposition (Protocol 07, item 7.5) shows that Novelty is the sole dimension where personas help (+0.154, $p_{\text{FDR}} = 0.001$) while Usefulness (−1.024) and Elaboration (−0.966) drive the reversal. The structure × persona interaction ($H_5$) is not significant ($F = 1.23$, $p = 0.27$).

> **Strategic Implication:** Pipeline structure is a lower-tier amplifier *under persona loading*, not a universal substitute: without personas, team structure lifts both tiers by the same small amount, and the case for cheap teams rests on their cost advantage (Section 4.1.4), not on amplified returns. Personas, by contrast, represent value destruction — they degrade quality while reducing cost only marginally.

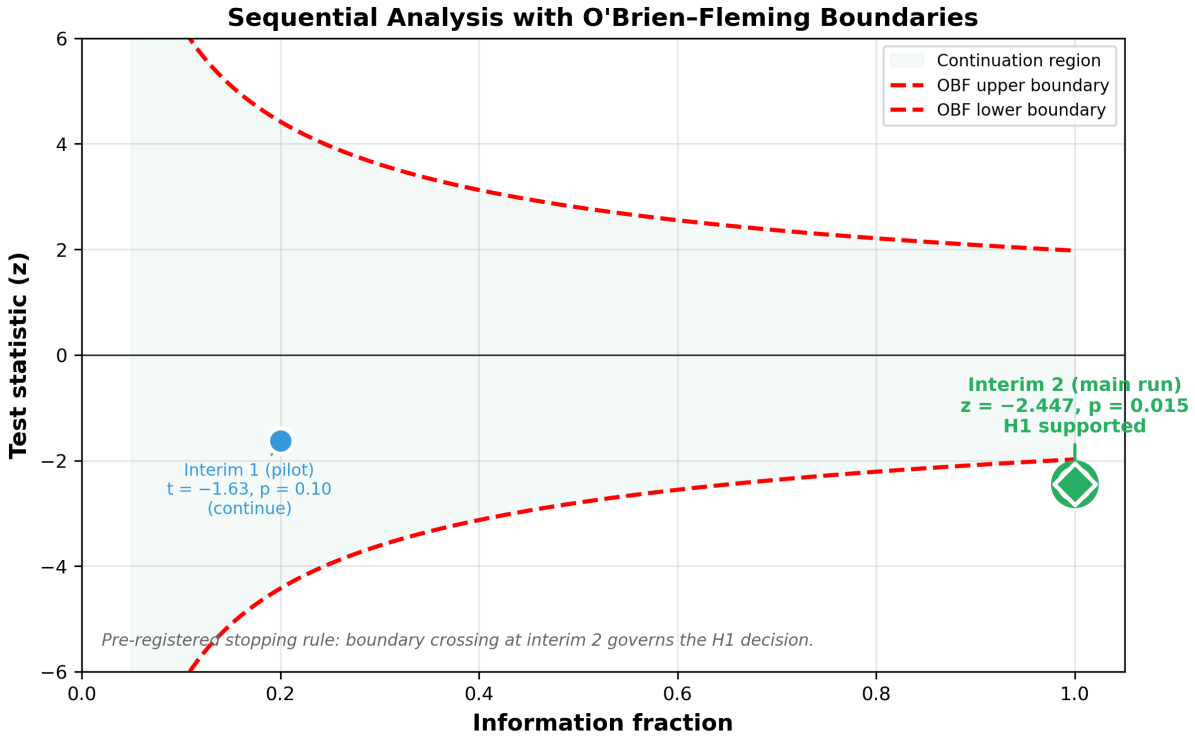



Figure 2: Sequential analysis with O'Brien–Fleming spending function. The test statistic $z = -2.447$ crosses the interim-2 boundary $|z| \geq 1.977$ (dashed line). The shaded region is the continuation region. The pre-registered decision is to stop and report $H_1$ as supported.

### 4.1.2 The persona penalty: style without substance

Every persona-loaded cell except the budget-extreme cheap solo is Pareto-dominated on the cost–quality plane (Section 4.1.4): C-T-P, F-S-P, and F-T-P are each beaten on quality by lower-cost no-persona cells. Personas do *not* raise cost — persona cells emit fewer tokens and price slightly *below* their no-persona twins — but the small saving buys a large quality loss, so the twin comparison on quality is uniformly negative ($d = -0.691$). The lone survivor, C-S-P, remains on the frontier only because it is the cheapest configuration tested; its quality is the lowest of all eight cells. Its headline efficiency ratio (T-EGR 8.60, the highest in the study) is purely a small-denominator artifact of that minimal cost — it marks the budget floor, not a recommended configuration, since the same tier without a persona (C-S-N) buys 0.30 points of quality for an additional \$0.005. The penalty is largest on dimensions that rubric readers associate with substance (Usefulness, Elaboration), smallest on the one most susceptible to stylistic signaling (Novelty). This pattern mirrors the contingency finding in the author's prior work (Hariri, 2026b) and is consistent with the jagged-frontier mechanism (Dell'Acqua et al., 2023): personas push the generator toward a more stylized region of its output space that the judge, correctly, penalizes on substance.

> **Strategic Implication:** Personas degrade output quality sharply while reducing cost only marginally — they are value-destroying for knowledge-work tasks. Firms should treat them as a failed investment rather than a quality lever.

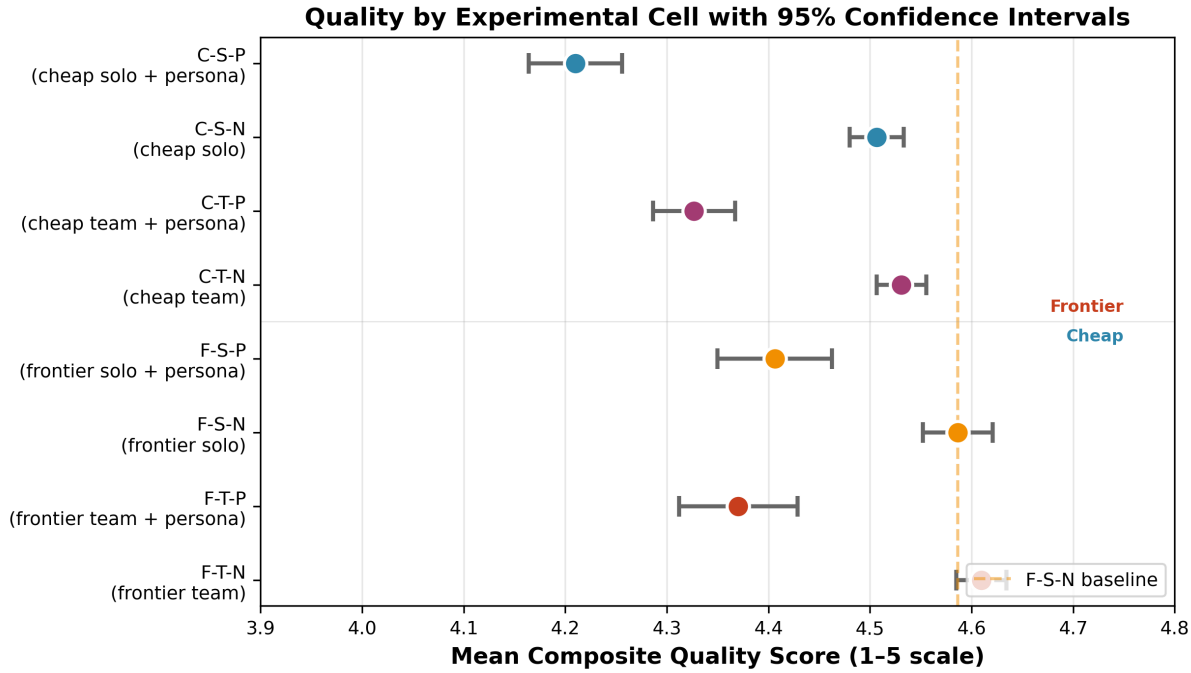



Figure 3: Quality by Experimental Cell with 95% Confidence Intervals.

#### 4.1.3 The frontier gap: lower-tier teams do not beat frontier solo

Paired comparisons between cheaper-team-with-persona (CTP) and the frontier-solo-no-persona baseline (F-S-N), matched on problem within each family (`results/ctp_vs_fsn_by_family.js` are negative and large in three families (qwen $d = -2.14$, gemini $d = -2.07$, claude $d = -1.32$; all $p_{\text{Holm}} < .001$, bootstrap CIs exclude zero), small and borderline in kimi ($d = -0.21$, $p_{\text{Holm}} = .053$), and null in gpt ($d = +0.03$, $p_{\text{Holm}} = .71$). The residual gap is thus family-dependent — consistent with the family-specificity findings throughout — and it is the empirical target that Study 2 (Section 4.2) asks a cheap refinement pass to close.

In business terms, the gap is small even where it is real. Judged on the 1–5 composite scale, the cheaper team with persona delivers 100.1% of its frontier solo sibling's quality for gpt, 98.1% for kimi, 95.8% for claude, 89.9% for qwen, and 87.6% for gemini (pooled: 94.3%) — and the persona-free cheaper team (C-T-N) actually *exceeds* its frontier sibling in two families (kimi 102.3%, qwen 100.6%). A budget owner should read the paired deficits above as statistical reliability, not as commercially visible differences: no family falls below 87% of frontier quality, at 86% of the frontier price.

> **Strategic Implication:** As executed, lower-tier teams do not beat a single frontier call on matched problems. The substitution thesis is bounded: structure improves lower-tier models but cannot fully close the frontier gap without additional investment.

#### 4.1.4 The Pareto frontier: optimal configurations for budget-constrained firms

The cost–quality map is shown in Fig. 6. At the primary $\varphi = 0.50$, the cell means are:

The Pareto frontier is **identical across all three** $\varphi$: {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N}. The dominated set {C-T-P, F-S-P, F-T-P} contains every persona-loaded team

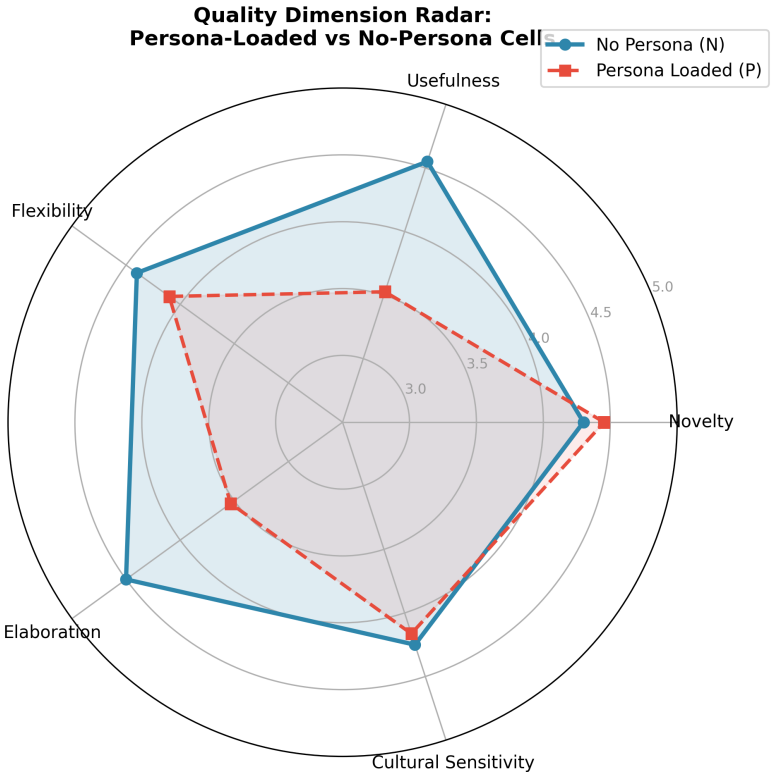



Figure 4: Quality-dimension radar for persona-loaded (red, dashed) versus no-persona (blue, solid) cells. The persona polygon is visibly smaller on Usefulness and Elaboration, slightly larger on Novelty — the visual signature of the style-over-substance tradeoff.

cell — the economic echo of the Section 4.1.2 persona penalty. Frontier membership is stable across the sensitivity panel, satisfying the pre-registered claim rule.

ICER (vs. the lowest-cost cell, C-S-P): C-S-N is the best buy at +$0.016 per point of composite quality; F-T-P is the worst at +$1.81 per point. The refined kimi C-T-P point (not in the table) sits at $0.0654 / quality 4.585 — 99.97% of pooled F-S-N quality (4.586) at 78% of its cost, with the caveat that the paired within-family comparison remains negative (disclosure per Section 4.2.1).

> **Strategic Implication:** The economically optimal strategy is tier substitution at the lower-tier end: a single call from a cheaper model (C-S-N) delivers 98.3% of frontier quality at 23% of the cost, and the three-step team variant (C-T-N) reaches 98.8% of frontier quality while buying coverage at 88% of the cost. The worst dollar in the experiment is spent on frontier teams — they cost 3.5× the frontier-solo price for a 100.5%-quality output, a gain of half a percentage point.

### 4.1.5 Robustness across model families and problems

The pre-registered robustness battery (Protocol 06) supports the Section 4.1.1 decision. Leave-one-family-out (item 6.5): $d$ ranges from −0.143 (drop kimi) to −0.303 (drop gemini), sign stable in all five folds; no single family drives the interaction. Leave-one-problem-

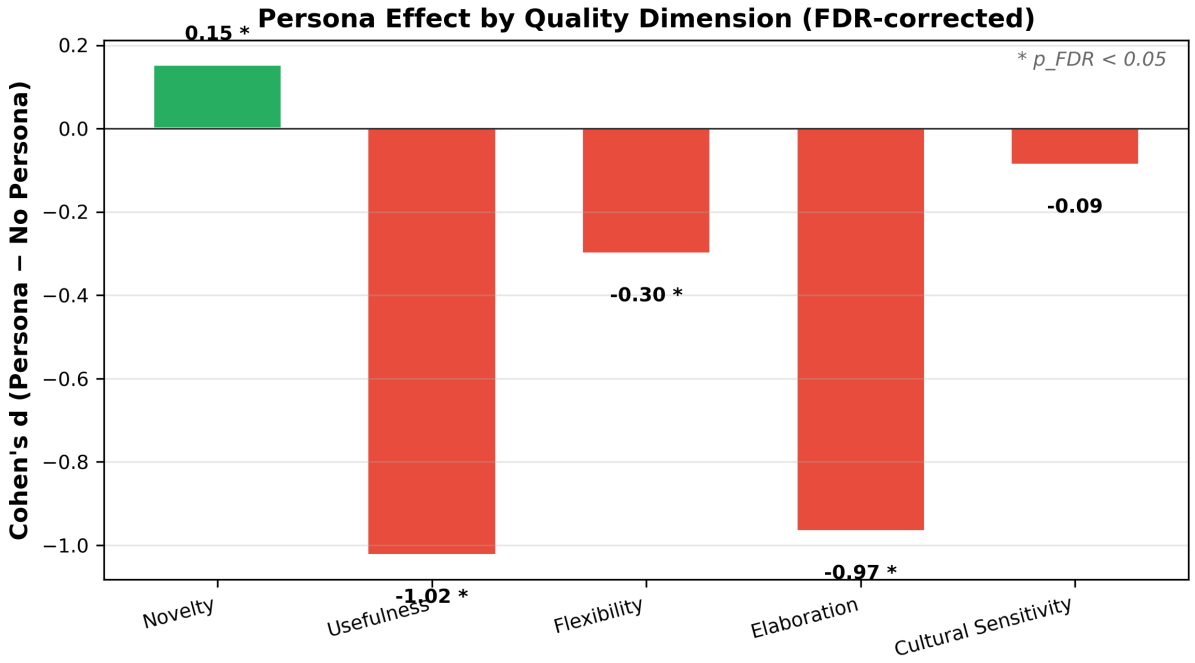



Figure 5: Persona Effect by Quality Dimension (FDR-Corrected).

Table 3: Cell summary statistics (primary analysis, $\varphi = 0.50$).

| Cell | $n$ | Cost (USD) | Quality | Quality (% of F-S-N) | T-EGR |
|---|---|---|---|---|---|
| C-S-P | 285 | 0.0145 | 4.210 | 91.8% | 8.60 |
| C-S-N | 285 | 0.0194 | 4.507 | 98.3% | 6.15 |
| C-T-P | 285 | 0.0715 | 4.327 | 94.3% | 1.58 |
| C-T-N | 285 | 0.0734 | 4.531 | 98.8% | 1.49 |
| F-S-P | 190 | 0.0670 | 4.406 | 96.1% | 1.36 |
| F-S-N | 190 | 0.0835 | 4.586 | 100% (base) | 1.00 (base) |
| F-T-N | 190 | 0.2951 | 4.610 | 100.5% | 0.27 |
| F-T-P | 190 | 0.3044 | 4.370 | 95.3% | 0.27 |

out (item 6.6): $d$ ranges from $-0.212$ to $-0.247$ across all 19 folds; no problem drives the interaction. Sensitivity power at the smallest cell ($n = 190$) detects $d = 0.288$ at 80% power; the observed $|d| = 0.227$ is detected through the within-subjects contrast over all 1,900 runs rather than per-cell power (item 6.7). The Type-II ANOVA sensitivity layer (item 6.3) agrees: tier × structure interaction $F = 6.90$, $p = 0.0087$, partial $\eta^2 = 0.0036$; residuals are non-normal, so the bootstrap intervals and Welch degrees of freedom remain the authoritative inference, as pre-specified. Length adjustment on the $H_1$ contrast yields $p = 0.63$ — the interaction is not a length confound. The random-effects meta-analysis over the five families (exploratory, Protocol 07, item 7.4) gives a pooled $d = -0.270$ (CI $[-0.472, -0.069]$, $z = -2.64$, $p = 0.008$) with low-moderate heterogeneity ($\tau^2 = 0.015$, $I^2 = 28.5\%$), confirming the interaction generalizes. Three further Protocol-07 checks are consistent with this reading. A mixed-model refit including persona as a factor (item 7.1) finds the tier × structure term null at the no-persona reference ($-0.001$, $p = 0.967$) and a significant three-way interaction with persona ($-0.151$, $p < 0.001$) — the moderation disclosed in Section 4.1.1. A TOST equivalence test on the no-persona structure effect (item

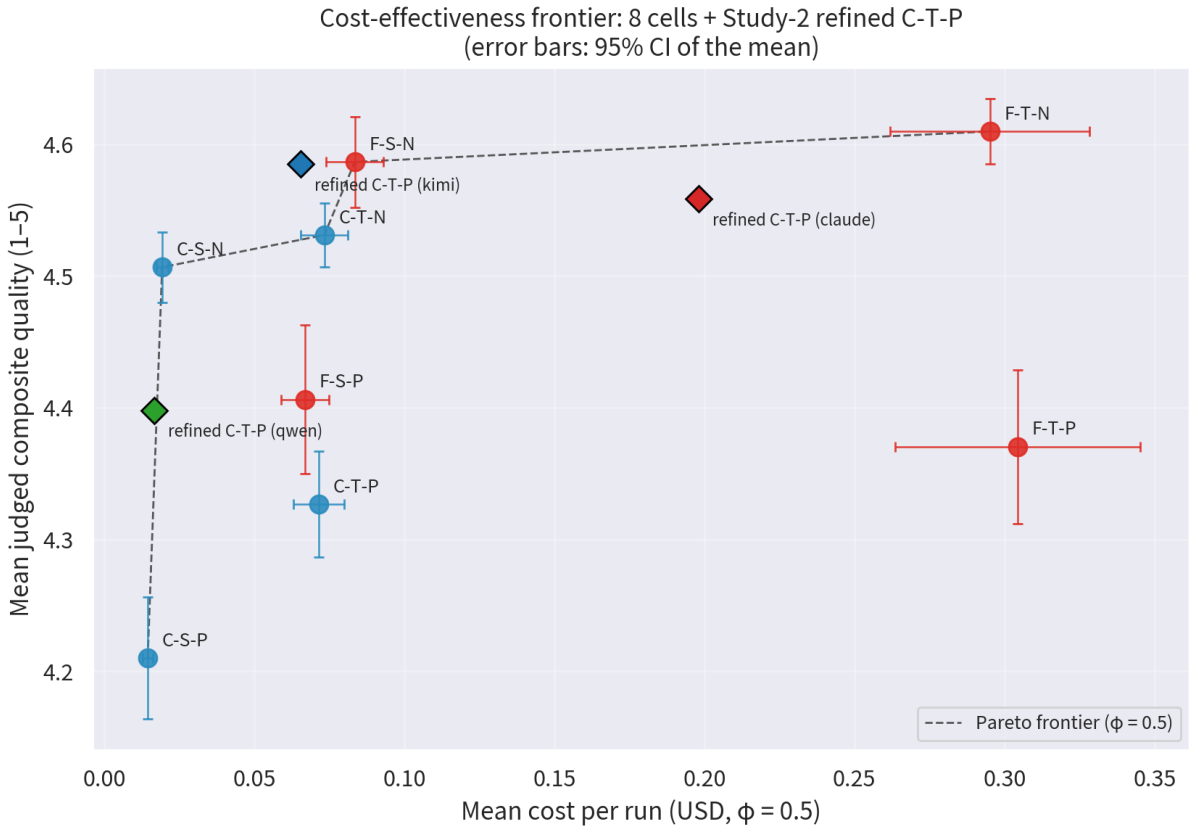



Figure 6: Cost-Effectiveness Frontier: 8 Cells + Study-2 Refined CTP (error bars: 95% CI of the mean).

7.2) is inconclusive ($d = -0.008$; equivalence not established, TOST $p = 0.063$ against $\pm 0.2$ bounds): the small structure effect cannot be certified as zero. Empirical-Bayes partial pooling of the per-family interaction estimates (item 7.3) shrinks all five toward the pooled value without sign flips (range $-0.20$ to $-0.36$).

Table 4 restates the battery in quality terms. With a pooled SD of 0.339 on the 1–5 composite and an F-S-N mean of 4.586, the observed interaction ($d = -0.227$) equals a 0.077-point structural gain for cheaper models — 1.7% of frontier quality — and every perturbation keeps the effect inside a 1.1–2.2% band.

Table 4: Robustness battery in business terms: effect sizes converted to quality points and percentage of frontier (F-S-N) quality.

| **Perturbation** | $d$ | **Quality points** | **% of frontier quality** |
|---|---|---|---|
| Primary interaction (all 1,900 runs) | $-0.227$ | 0.077 | 1.7% |
| Leave-one-family-out range | $-0.143$ to $-0.303$ | 0.048–0.103 | 1.1–2.2% |
| Leave-one-problem-out range | $-0.212$ to $-0.247$ | 0.072–0.084 | 1.6–1.8% |
| Cross-family meta-analysis (pooled) | $-0.270$ | 0.091 | 2.0% |

Across all battery perturbations the interaction keeps its sign and the decision is unchanged. The leave-one-out sensitivity (Fig. 9) and random-effects meta-analysis (Fig. 10) confirm that no single family, problem, or outlier drives the result.

**Strategic Implication:** The pattern is not a vendor or problem-set artifact:

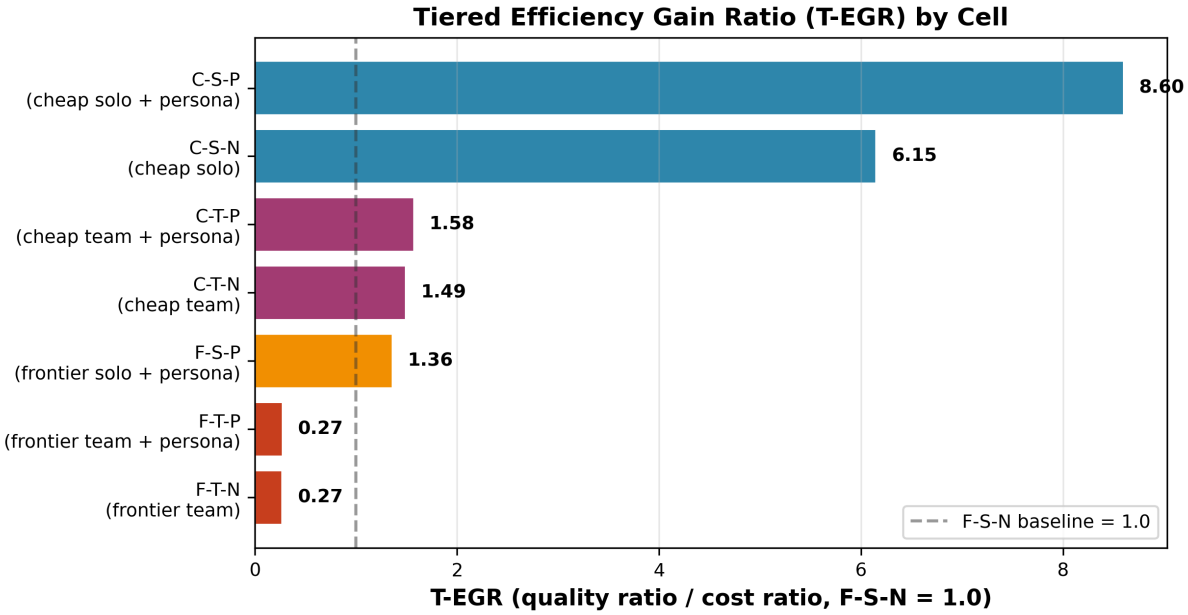



Figure 7: Tiered Efficiency Gain Ratio (T-EGR) by Cell.

remove any single model family or any single problem and the conclusion is unchanged. In business terms, the structural lift for cheaper models is a reliable 1–2% of frontier quality wherever you look — modest in absolute terms, but stable enough that a firm can adopt the lower-tier strategy without betting on any particular vendor, and small enough that no vendor switch will overturn it.

#### 4.1.6 Exploratory heterogeneity: domain contingency, a capability-gap mechanism, and the persona decomposition

**Status declaration.** Everything in this subsection is *exploratory and post-hoc*: it was formulated after data lock, is not part of the pre-registered hypothesis set $H_1$–$H_6$, and consists exclusively of re-analyses of the frozen Study-1 artifacts (no new generation or judging calls). We report it because it generates concrete, falsifiable predictions — which were then tested by the pre-registered confirmatory replication wave (Section 4.1.7). Anticipating that section: **all three upgrade candidates below (domain contingency, the capability-gap mechanism, the marketing variance advantage) FAILED their binding out-of-sample gates** and therefore remain exploratory; the numbers in this subsection stand as the Study-1 record, and the wave numbers are reported alongside in Section 4.1.7 (Nosek et al., 2018; Open Science Collaboration, 2015).

**Where does heterogeneity live?** Three screening analyses localize the variation. Problem difficulty does not moderate the team effect ($r = 0.153$, $p = 0.53$ across the 19 problems). Family quality rankings are remarkably stable across problem domains (Kendall's $W = 0.922$; Kendall & Babington Smith, 1939), so family choice is largely domain-invariant. But measurement reliability itself is domain-heterogeneous: single-judge agreement ICC(3,1) (Shrout & Fleiss, 1979) ranges from 0.235 (HR/organization) to 0.523 (marketing), with panel reliability ICC(3,9) between 0.73 and 0.91 — HR answers are simply harder for any judge, human or LLM, to score consistently.

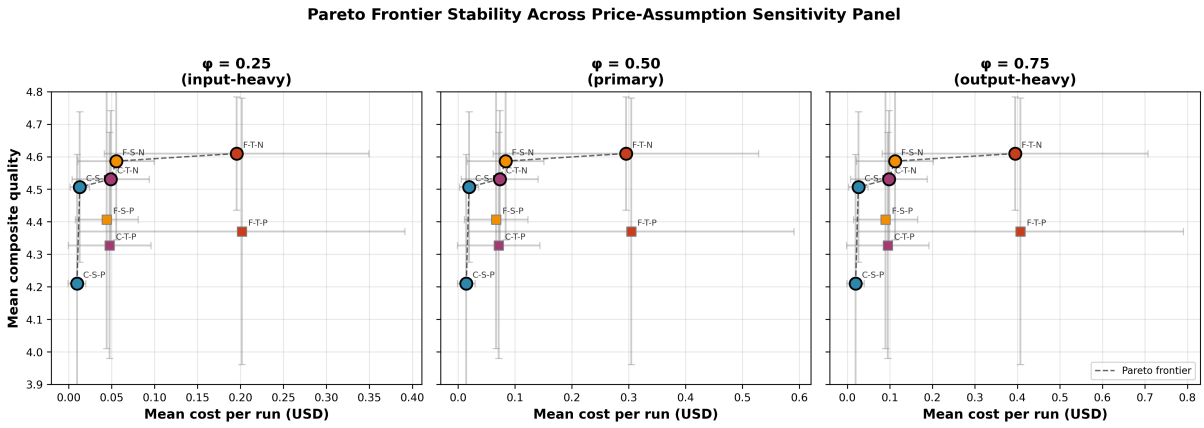



Figure 8: Pareto frontier stability across the price-assumption sensitivity panel. Left: $\varphi = 0.25$ (input-heavy blend). Center: $\varphi = 0.50$ (primary analysis). Right: $\varphi = 0.75$ (output-heavy blend). The frontier set {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N} is identical in all three panels, satisfying the pre-registered cross-$\varphi$ stability rule.

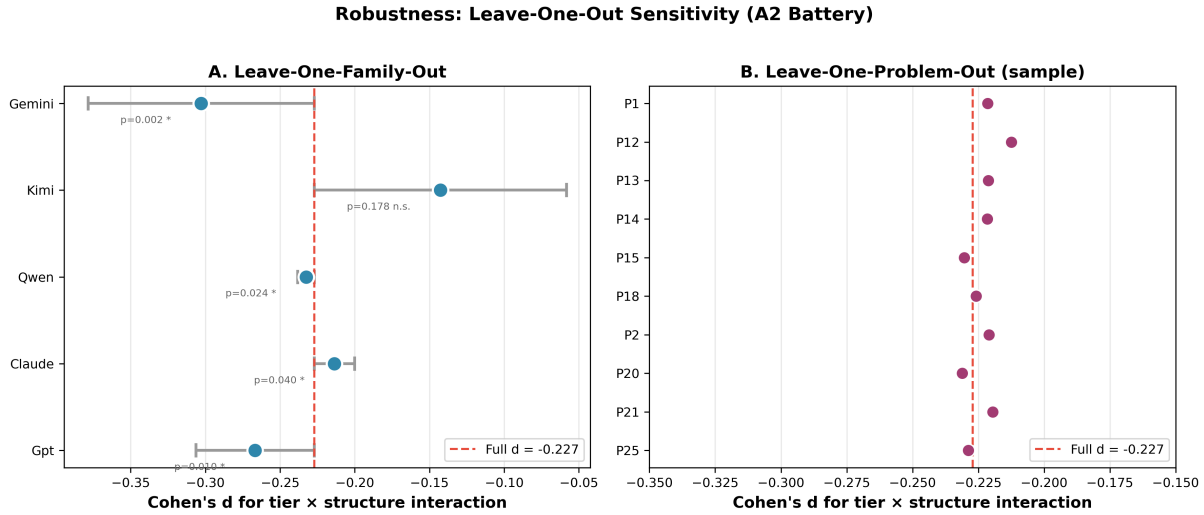



Figure 9: Robustness: Leave-One-Out Sensitivity (A2 Battery).

**The team effect is domain-contingent.** Pooling the cheap tier, the team-minus-solo quality effect varies from +0.004 (marketing — the only domain where teams match solo) through strategy (−0.036), HR (−0.053), operations (−0.058), product innovation (−0.103), to finance (−0.106). In marketing the team advantage extends beyond the mean: team output variance is significantly *lower* than solo (Brown–Forsythe test, $p = 0.020$; Brown & Forsythe, 1974), and downside risk shrinks across cells ($P(\text{composite} < 4.0)$: solo 2.8%, team 1.8%, frontier 1.1%). The ensemble-variance logic is classical — combining models reduces error when members are accurate and diverse (Hansen & Salamon, 1990; Dietterich, 2000) — but here it appears only for communication-intensive tasks.

**A capability-gap mechanism.** The family-level team effect (team minus frontier solo) correlates almost perfectly with the family's capability gap (frontier minus cheap solo) (Pearson $r = -0.978$, $p = 0.004$; Spearman $\rho = -1.000$; $n = 5$ families): the team effect is most positive where the cheap model already matches its frontier sibling (kimi, −0.164 gap — the cheap model scores *above* its frontier — and +0.105 team effect)

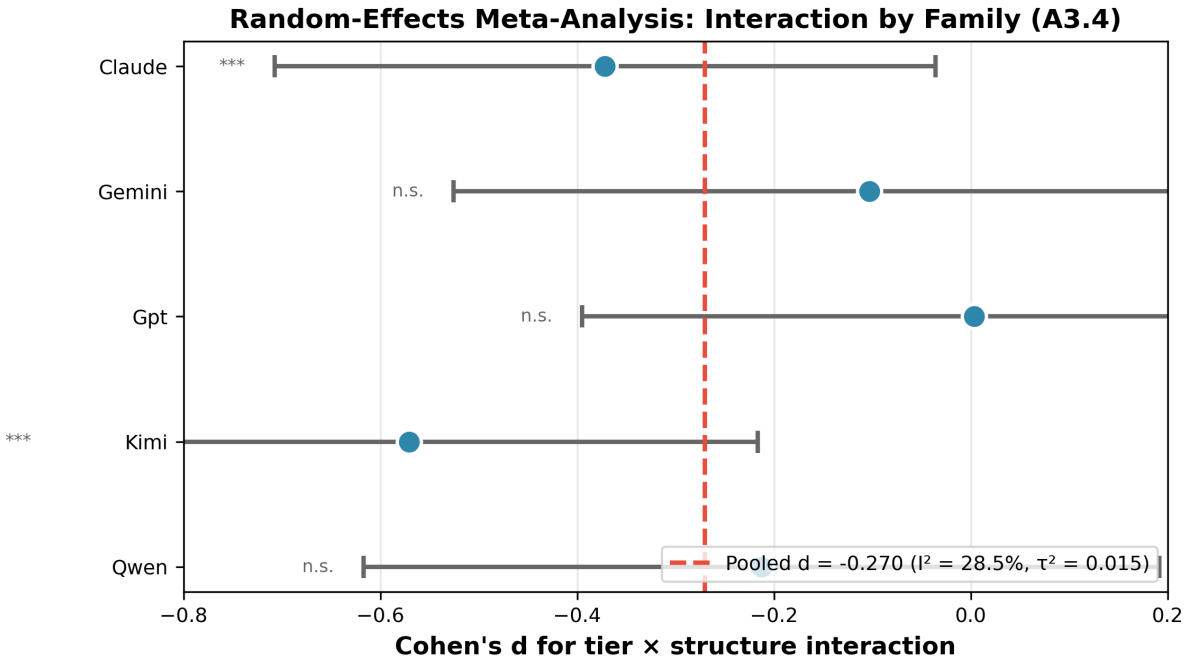



Figure 10: Random-Effects Meta-Analysis: Interaction by Family.

and most negative where the cheap model lags farthest below it (claude, +0.364 gap, −0.292 team effect). One caution applies beyond the small $n$: the correlation is partly mechanical, because the team effect and the gap both contain the frontier mean. With only five families this remains a hypothesis generator, not a test; the replication wave expands the family universe specifically to power this correlation properly (moderator analysis logic as in Borenstein et al., 2009).

**Length control reveals an asymmetry.** A length-adjusted mixed model (composite ∼ log-tokens + cell, random family and problem intercepts; Westfall et al., 2014) leaves the frontier advantage intact (+0.078, $p < 0.001$) but reduces the team's small raw edge (+0.026) to −0.018 ($p = 0.40$). We deliberately do *not* read this as mediation: quality is judged on visible full text, and team structure mechanically produces longer visible answers, so controlling length partially controls the intervention itself. The defensible reading is the asymmetry — frontier superiority survives length adjustment; the team edge does not. The same model locates the remaining economics: frontier solo is *faster* than cheap teams (94.8s vs. 131.7s) and cheap solo dominates efficiency (295.8 vs. 28.1 quality points per dollar-minute for teams).

**Persona decomposition.** Extending the battery to the persona axis (full $2 \times 2 \times 2$ coverage): (i) personas produce no spillover into the judged Cultural Sensitivity dimension (pooled $d = -0.10$ to $-0.01$, all domain cells n.s.; the personas were designed as a general cultural-perspective injection, not as an instrument for this specific rubric item); (ii) the solo-persona novelty gain is genuine — it survives length control (+0.465, $p < 0.001$) and is uniform across all six domains (+0.36 to +0.57) against a uniform usefulness cost (−0.96 to −1.10); (iii) teams destroy that gain (structure × persona interactions: Novelty −0.693, Usefulness +0.711, both $p < 0.001$, Holm-corrected; Holm, 1979); (iv) persona answers are judged *more* consistently (panel ICC(3,9) 0.933 vs. 0.918), so the effect is not a measurement artifact; and (v) personas roughly double outcome variance (solo SD 0.232 → 0.397), consistent with a style-injection account. Length bias in LLM judging is

a documented threat (Zheng et al., 2023; Dubois et al., 2024), which is why each effect above was re-tested under length control before reporting.

> **Strategic Implication (exploratory — did not survive replication):** Treat everything in this subsection as an untested lead, not a budgeting rule. In our first sample, marketing looked like the one domain where cheap teams matched solo output *with lower variance* — an attractive niche for volume automation. On fresh data that niche disappeared, and the "capability-gap" shortcut for predicting where teams help also evaporated. Practical reading: do *not* route work by domain and do *not* expect team structures to stabilize quality; base budget decisions only on the replicated rules in the decision framework of Section 5.2.

#### 4.1.7 Out-of-sample confirmation: the replication wave (Protocol 12)

**Design.** Protocol 12 (pre-registered, OSF jg49x, before any wave generation) froze a binding upgrade rule: a Study-1 *exploratory* finding becomes a headline claim only if its pre-registered wave test passes at the pre-registered alpha and direction; the core claims are re-tested with the same frozen instruments. The wave uses 30 fresh post-cutoff business problems (a second $6 \times 5$ domain grid, zero overlap with Study 1), the three core cells (C-S-N, C-T-N, F-S-N), and a family universe doubled from 5 to 10 (adding Mistral, MiniMax, Gemma, GPT-OSS, and ThinkingMachines): 2,400 generation runs, judged by the same frozen three-judge panel under the same three verbatim prompt variants (21,762 judgment cells; 2,418 analyzed runs). The analysis script was frozen and hash-locked before judging; a disclosed numerical fallback (a singular dual variance-component LMM, replaced under a pre-declared rule that hard-requires bit-identical fixed effects) is recorded in the results artifacts and our decision log (D52). Provider events are disclosed, not imputed: the Gemini lane was deferred on a free-tier daily cap until a billing upgrade, and the Kimi frontier lane was completed via a Together.ai routing deviation (same pinned model version; D53) after Moonshot credit exhaustion.

**Gate outcomes.** Table 5 reports the complete-panel results (the interim 9-family record of 2026-08-24 stands unchanged in the wave report; the complete panel adds Gemini and the Kimi remainder).

**Reading.** The three core claims replicate out-of-sample on the full family universe: (i) cheap teams beat cheap solo by a small but real margin (W1); (ii) frontier solo beats cheap teams decisively on quality (W2, an effect $\sim 4.4\times$ the team effect); (iii) the cost–quality frontier is bimodal — cheap solo for volume economics (T-EGR = 4.04), frontier solo for quality — with cheap teams the dominated middle at current list prices (W6; the ordering is unchanged under the MiniMax promotional-price sensitivity vector, T-EGR 4.11/0.98). All three exploratory upgrade candidates fail: the "marketing exception" was a sample-specific fluctuation (marketing flips negative, $\Delta = -0.075$), the near-perfect capability-gap correlation was an $n = 5$ artifact, and teams do not reduce downside variance in any domain. We read the failed gates as the design working as intended: the pre-registration separated what survives contact with new data from what does not (Open Science Collaboration, 2015; Camerer et al., 2018), and no post-hoc relabeling is permitted in either direction.

> **Key Takeaway:** the headline economics of this paper are replication-grade; the contingency nuances are not. Practitioner rules should be built on W1/W2/W6 only.

Table 5: Confirmatory replication wave: gate outcomes on the complete 10-family panel ($n = 2{,}418$ judged runs, $\alpha = 0.05$).

| Gate | Result (complete panel) | Verdict |
|---|---|---|
| W1 team effect (C-S-N → C-T-N) | +0.0168, $p = 0.002$ | PASS |
| W2 upgrade threshold (C-T-N → F-S-N) | +0.0745, $p < 0.0001$ | PASS |
| W3 domain contingency | interaction LRT $p = 1.0$; directional signs 5/6; *no* positive domain (marketing flips negative) | FAIL |
| W4 capability-gap mechanism | $r = -0.214$, $p = 0.55$ ($n = 10$ families; power for $\rho = 0.80$ is 0.83) vs. Study-1 $r = -0.978$ | FAIL |
| W5 marketing variance advantage | Brown–Forsythe $p = 0.795$, direction reversed ($SD_{team}$ 0.272 > $SD_{solo}$ 0.263) | FAIL |
| W6 economics | F-S-N 40.6 > C-T-N 21.2 quality/$-min; Pareto = {C-S-N, F-S-N}; T-EGR 4.04 (C-S-N), 0.96 (C-T-N) | PASS |
| W7 judge reliability (audit) | ICC(3,1) 0.471–0.622 by domain | — |

**Strategic Implication:** The rules you can take to the board are the three that survived contact with new data. (i) Adding pipeline structure to a cheap model buys a small but real quality gain. (ii) Frontier solo still wins on quality by a margin $\sim 4\times$ larger — for revenue- or compliance-bearing tasks, pay for the frontier. (iii) The efficient menu is two-point: cheap solo for volume work (best quality per dollar, T-EGR 4.04), frontier solo for high-stakes work — cheap teams sit in the dominated middle at current prices. Every other pattern in this paper is a lead for future research, not an operating rule.

## 4.2 Study 2: Why Substitution Stops Short of the Frontier

Study 1 established that orchestrated lower-tier models approach but do not reach the frontier (Section 4.1.3). Study 2 asks the follow-up a budget owner actually faces: *can a cheap refinement pass close that residual gap — and if it cannot, why not?* The answer is what separates a risk-mitigation strategy from a replacement strategy.

### 4.2.1 The refinement ceiling: closing the gap at low marginal cost

A frozen, add-only refiner is applied to the CTP outputs. The pilot satisfied the feasibility gate (G1, 100% completion); the locked pool comprised all 171 matched CTP–FSN pairs (57 per family), of which the pilot covered 20 per family and a pre-registered extension (registered before execution) covered the remainder. The pre-registered superiority gate (G2: $d \geq 0.2$ with CI excluding 0 in all three families, in the direction refined > FSN) is mechanically failed in all three families: kimi $d = -0.295$, claude $d = -1.301$, qwen $d = -0.809$ (all CIs strictly negative after Holm). The superiority claim is formally rejected.

Refinement nevertheless does repair work where a real gap exists. Measured as gap recovery against each family's own pre-refinement CTP-vs-FSN gap (Section 4.1.3), the pass recovers 64% of the gap for qwen — the family with a large pre-refinement gap. For kimi the pre-refinement gap is already near zero (mean difference −0.086), so recovery is small on the mean (14%) and undefined in effect-size terms: the refined output's reduced variance makes the paired $d$ slightly *larger* (−0.295 vs. −0.21) even though the mean moves toward F-S-N — refinement cannot close a gap that is not there. For claude the mean recovery is −3% (no movement). The Novelty protection criterion (Protocol 09 criterion H2.2: no family loses more than −0.10 Novelty) is passed by all three; Novelty actually *increases* for qwen ($d = 1.10$, $p_{\text{FDR}} < 0.001$). Usefulness and Elaboration show large gains in kimi and qwen ($d = 0.57$ to 1.35, all $p_{\text{FDR}} < 0.001$) and are unchanged in claude — which was already at ceiling on those dimensions before refinement.

Length analysis confirms the dilution hypothesis is *not* the driver: the pooled Pearson correlation between word-count ratio (refined/original) and composite-score change is $r = +0.67$, Spearman $\rho = +0.70$ ($p < 0.001$). Longer refined answers score higher, not lower.

Cross-family comparisons (Fig. 12) remain negative after refinement: no cross-family CTP pairing flips to superiority, confirming that the within-family result is not an artifact of family selection.

> **Strategic Implication:** A refinement pass costing \$0.001–0.003 per output (family-dependent; pooled mean ≈ \$0.002) delivers bounded improvement — it recovers most of the frontier gap on average but never achieves paired superiority. For firms, this means refinement is a viable risk-mitigation strategy (improve quality at lower cost) but not a replacement strategy (it will not beat the frontier).

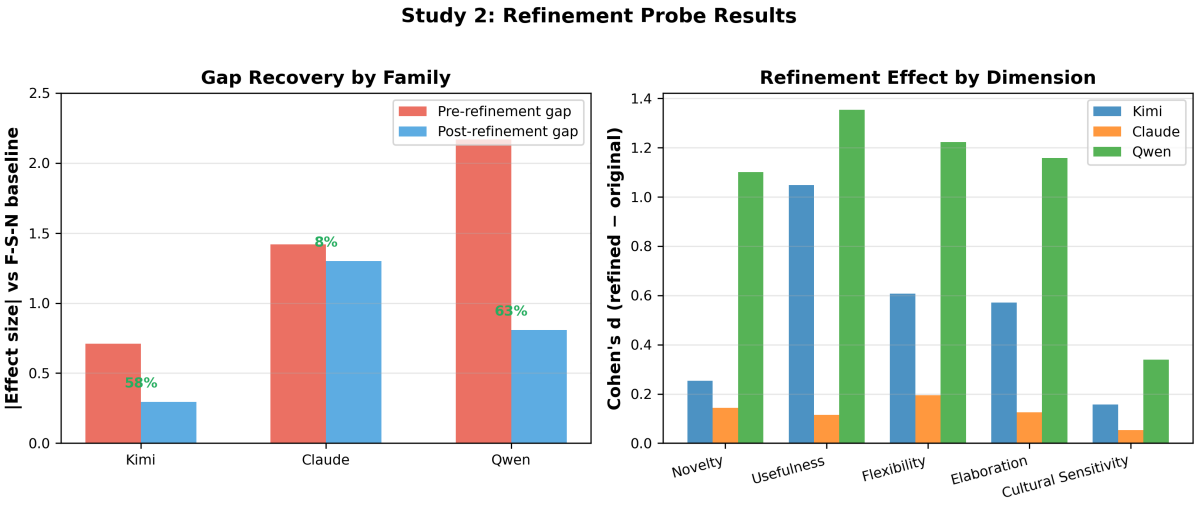



Figure 11: Study 2: Refinement Probe Results.

### 4.2.2 Why orchestration buys coverage but not density

Thirty triples (10 per family) are coded under a 13-code codebook. Pooled Cohen's $\kappa = 0.824$ (93.1% agreement), well above the 0.70 gate. Six of the thirteen codes fall

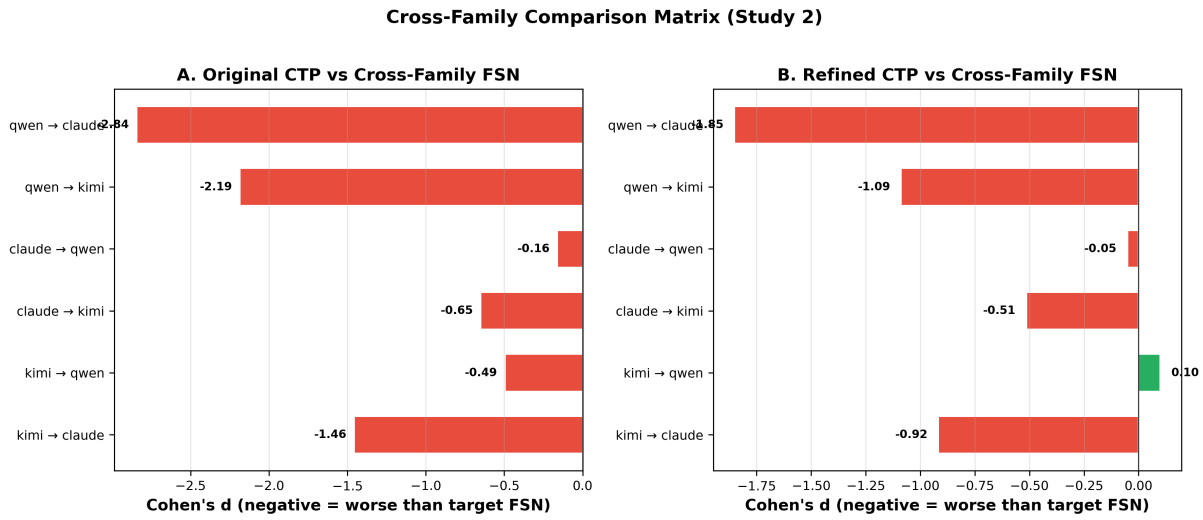



Figure 12: Cross-Family Comparison Matrix (Study 2).

below 0.70 (`diagnostic_reframe` 0.54, `named_concepts` 0.49, `assumption_test` 0.56, `structured_report` 0.59, `external_authority` 0.56, `redundancy_bloat` 0.65), a base-rate artifact: percentage agreement on these codes ranges from 82% to 98%, but $\kappa$ is depressed by near-zero prevalence. Conclusions on below-threshold codes are disclosed as medium reliability.

Of the six codes whose prevalence changed between original and refined, the pattern differs by family. For kimi, all six changed codes are *structural scaffolding* codes: `operational_parameters`, `structured_report`, `phased_timeline`, `tables_matrices`, `quantified_targets`, `granular_ops_detail`. For qwen, five of the six are structural scaffolding (the same list minus `structured_report`) and one — `diagnostic_reframe` — is an insight code. No other insight code (`named_concepts`, `assumption_test`, `concise_maxims`) moved in either family. The convergence metric compares, for each of the 13 codebook codes, how far the original and refined lower-tier answers sit from the frontier answer's profile: a code counts as "toward" if refinement reduces that distance and "away" if it increases it. Refinement does not consistently move lower-tier outputs toward the frontier (kimi: toward 4 / away 6, net divergence; qwen: 5/4; claude: 0/2 with 11 codes flat — claude's originals were already at the codebook ceiling). Where outputs move away, refinement is adding structure the frontier answers do not carry — overshooting on coverage rather than gaining insight density.

The residual frontier edge is precisely located. The only $\kappa$-reliable frontier-vs-refined difference $\geq 0.30$ is `concise_maxims`: qwen frontier 1.00 vs. refined 0.55 (gap 0.45). The medium-reliability `redundancy_bloat` code is 0.00 in every frontier sample and 0.65–0.95 in every refined sample. Claude's original answers are already at the codebook ceiling ($\geq$ 0.90 on 11 of 13 codes); the number of changed codes between original and refined is zero — a saturation mechanism that explains the absent gap recovery ($-3\%$) in Section 4.2.1.

> **Strategic Implication:** Orchestration is a coverage play — it adds structure, length, and completeness. The frontier's advantage is a density play — compressed insight with zero waste. Add-only refinement cannot manufacture density, which explains why lower-tier teams approach but do not surpass frontier quality.

**A note on replication scope.** Unlike Study 1 and Study 3, whose headline claims were each stress-tested on fresh problems in a pre-registered replication (Section 4.1.7, Section 4.3.1), Study 2 was not carried into a confirmatory wave. Its two findings therefore rest on the original locked sample: the refinement ceiling is a pre-registered gate result — the G2 superiority test failed mechanically in all three families (Section 4.2.1) — and the coverage-versus-density mechanism is a mixed-methods coding result ($\kappa = 0.824$; Section 4.2.2). We report Study 2 as a *mechanistic explanation* of the Study 1 ceiling: well-supported on its own sample, but not yet independently replicated. The natural next step — a *subtractive* refiner permitted to cut as well as add — is named in Section 8.

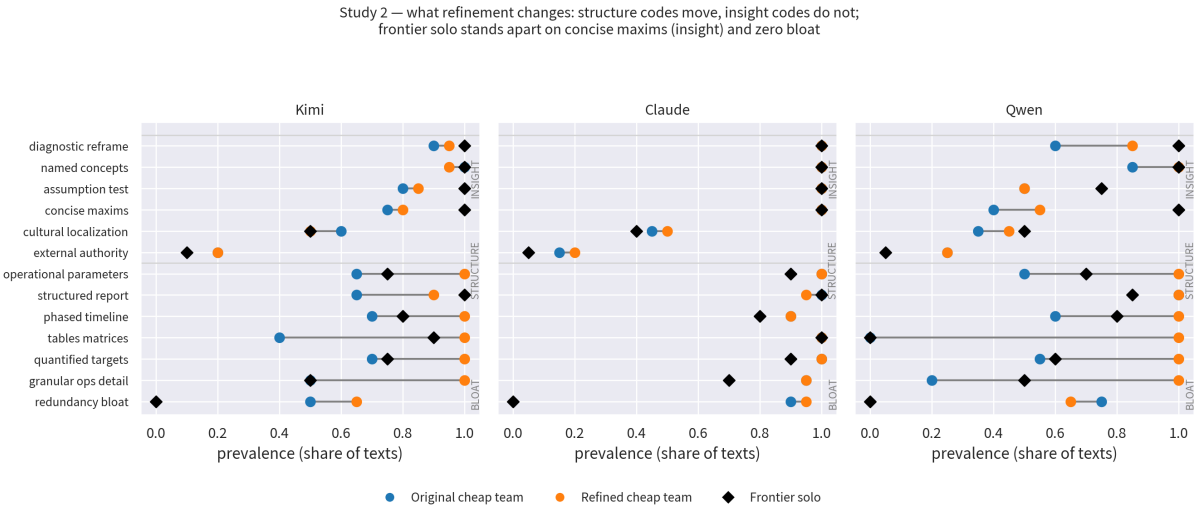



Figure 13: Code prevalence by family (Study 2): original cheap-team answer (blue), refined answer (orange), frontier solo (black diamond), for the 13 codebook codes grouped by research question. Refinement moves the *structure* codes rightward but leaves the *insight* codes unmoved; the frontier stands apart on concise maxims and zero bloat.

## 4.3 Study 3: Who Should Design the Pipeline?

**Design.** Every pipeline in Studies 1–2 was *human-designed*: the builder → critic → refiner structure was our fixed research instrument (the registered-but-unexecuted A7 arm isolated exactly this variable). Study 3 (Protocol 13; operative pre-registration OSF bwc93 — an initial v1 registration was superseded by this instrument revision and withdrawn — with attrition clarification 639eq; all registrations timestamped before the corresponding computation) executes A7 as a stand-alone study. Using DeeWaanAI's production auto-designer, the model receives only the problem text and authors a bespoke multi-step pipeline — free choice of roles, step count, and critic-gated if/while feedback loops — which is then executed verbatim by the cheap tier. Two arms per family: **D-SELF** (the cheap sibling designs *and* executes) and **D-FRONT** (the frontier flagship designs; the cheap sibling executes). The 19 confirmatory problems × 3 replications × 2 arms × 5 families = 570 planned runs; 544 completed, with 26 disclosed attrition events concentrated in the Kimi lane (21 designer-stage failures — the cheap designer returned empty or truncated JSON on the production design prompt — plus 4 execution failures and 1 Claude designer failure). The pre-registered gates frame this flow: feasibility gate G1 (per-family completion $\geq 80\%$ excluding moderation-blocked tasks) passed for four families and failed strictly for Kimi ($89/114 = 78\%$); per the amended G1 rule, the Kimi lane

is retained with the fully disclosed attrition above. The superiority-stop gate G2 (halt only if D-FRONT trails the frozen C-T-N baseline by $d < -0.30$ on Novelty) passed at pilot ($d = +0.42$), clearing the main run. Solutions were judged by the same frozen three-judge panel under the three verbatim prompt variants (4,896 unique judgments). The primary contrast (A) tests H-D1: D-FRONT vs. D-SELF on Novelty, paired within family × problem, Holm-adjusted across families; contrast B tests H-D2, both arms against the frozen human-designed C-T-N baseline; an author-requested contrast informs H-D3, testing both arms against the frozen frontier-solo (F-S-N) baseline, and a cost analysis prices every run at the frozen price vector.

**Contrast A: authorship barely matters.** Table 6 reports the primary contrast. Frontier-designed pipelines are statistically indistinguishable from self-designed ones in four of five families; the lone significant effect favors frontier authorship for Claude ($d = 0.78$, $p_{\text{Holm}} = 0.014$). Pooled across families, the authorship gap on Novelty is 4.03 vs. 3.99 — under 0.1 scale points. A post-hoc robustness check on the five-dimension composite (registered in the Protocol-13 plan) leaves the conclusion unchanged: Claude is the only family with a significant authorship effect on either DV (composite $d = 0.35$, $p_{\text{Holm}} = 0.021$); every other family is null on both.

Table 6: Study 3 primary contrast: D-FRONT vs. D-SELF on Novelty, paired within family × problem ($n = 19$ pairs), Holm-adjusted across the five families.

| **Family** | Δ (D-FRONT − D-SELF) | $t$ | $d$ | $p_{\text{raw}}$ | $p_{\text{Holm}}$ |
|---|---|---|---|---|---|
| Claude | +0.136 | 3.00 | 0.78 | 0.003 | **0.014** |
| GPT-5.6 | +0.049 | 1.77 | 0.22 | 0.078 | 0.310 |
| Qwen | +0.021 | 0.73 | 0.10 | 0.463 | 1.0 |
| Gemini | +0.021 | 0.55 | 0.08 | 0.582 | 1.0 |
| Kimi | −0.038 | −0.64 | −0.12 | 0.522 | 1.0 |

**Contrast B: both arms beat the human design.** Against the frozen human-designed C-T-N pipeline, model-authored pipelines win in *all five families on both arms* (D-SELF $d = 0.15$–$0.67$; D-FRONT $d = 0.24$–$1.33$ on Novelty). The auto-designer also discovers structure we did not hand it: 98.9% (D-SELF) and 100% (D-FRONT) of designs include at least one critic-gated if/while feedback loop, averaging 5.3–5.4 designed steps and 5.8–6.2 executed steps after gate loops — richer than our fixed three-step instrument. Cheap-designed pipelines consume more tokens overall (34.8M vs. 24.3M across the study), but the gap is execution-driven (34.0M vs. 23.6M): design-stage usage is comparable across arms (0.6M vs. 0.7M, 2–3% of totals) — cheap executors burn tokens in longer, looped executions, not in more design iteration.

**Contrast vs. frontier solo: orchestration is family-specific, not general.** Pooled, the arms are at parity with frontier solo (D-SELF $d = -0.22$; D-FRONT $d = -0.05$ on Novelty) — but the pooled figure hides the story (Table 7, Fig. 15). Cheap auto-designed pipelines *beat* their own family's frontier flagship running solo for Kimi ($d = +0.85/+0.74$, $p_{\text{Holm}} = 0.006/0.013$), Qwen ($d = +0.55/+0.61$), and Gemini ($d = +0.46/+0.51$, n.s. after Holm); they tie GPT-5.6-solo; and they lose catastrophically to Claude-opus solo ($d = -3.15/-2.84$). By domain, pooled D-FRONT means beat frontier solo in strategy (4.08 vs. 4.00) and marketing (4.08 vs. 4.01), tie in HR (4.11 vs. 4.10) and product (4.24 vs. 4.25), and lose in finance (3.81 vs. 3.99) and operations (3.83 vs. 3.95) — but a descriptive family × domain breakdown (cells of 2–5 problems, underpowered

for inference by design; Fig. 14) shows the domain pattern reversing sign within families: Claude loses in all six domains while Kimi and Gemini win in most. The moderator is the *family* (how high the solo frontier sits), not the task. Frontier authorship does not rescue the losing families: where the cheap executor is the bottleneck, a frontier-designed plan cannot compensate.

Table 7: Study 3 economics: cost per completed solution (USD, frozen price vector v2026-08-23) and Novelty effect vs. frontier solo (F-S-N). $\Delta Q$ = Cohen's $d$ on Novelty vs. F-S-N. Positive $\Delta Q$ at cost ratio $< 1$ = strictly dominant orchestration.

| **Family** | **F-S-N cost** | **D-SELF** | | **D-FRONT** | |
|---|---|---|---|---|---|
| | | cost (×F-S-N) | $\Delta Q$ | cost (×F-S-N) | $\Delta Q$ |
| Gemini | \$0.043 | \$0.010 (0.24×) | +0.46 | \$0.025 (0.59×) | +0.51 |
| Qwen | \$0.020 | \$0.040 (2.0×) | +0.55 | \$0.046 (2.3×) | +0.61 |
| Kimi | \$0.107 | \$0.377 (3.5×) | +0.85 | \$0.274 (2.6×) | +0.74 |
| GPT-5.6 | \$0.180 | \$0.729 (4.0×) | −0.07 | \$0.512 (2.8×) | +0.13 |
| Claude | \$0.266 | \$0.438 (1.7×) | −3.15 | \$0.668 (2.5×) | −2.84 |

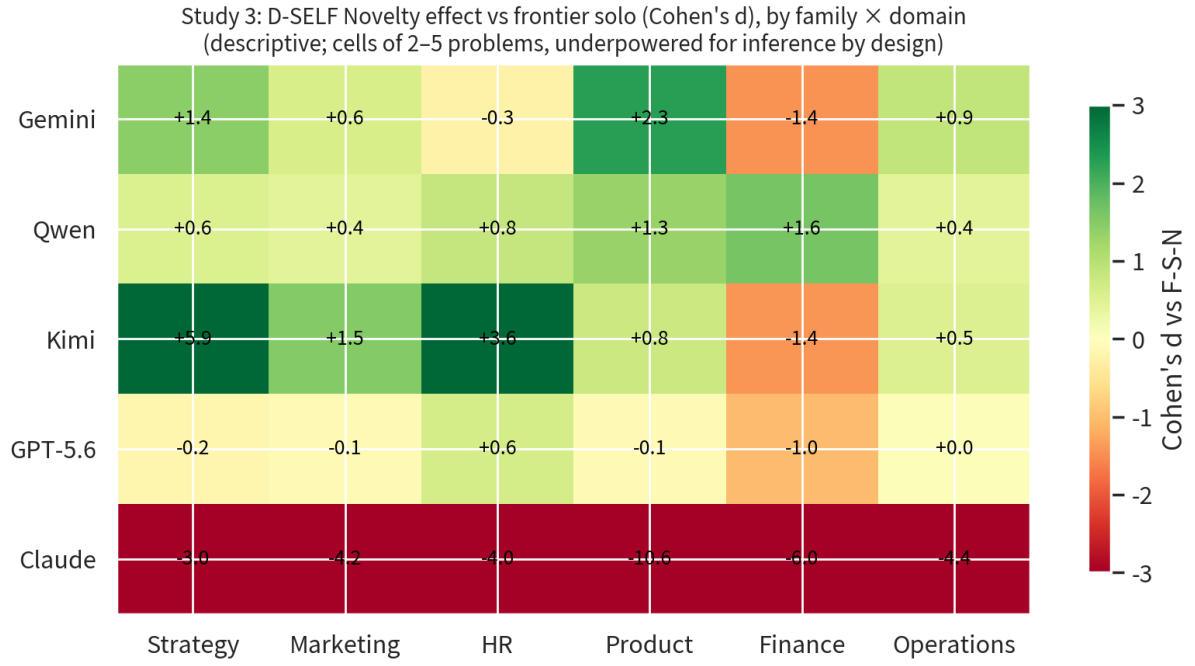



Figure 14: Study 3 family × domain landscape: the Novelty effect of cheap auto-designed pipelines versus frontier solo (F-S-N) across the five families and six business domains (descriptive; cells of 2–5 problems, underpowered for inference by design). The visible moderator is the *family* — how high its solo frontier ceiling sits — not the task domain (Section 4.3.1).

**Decision rule.** Study 3 converts orchestration from a blanket recommendation into a routing table. For Gemini, cheap auto-designed pipelines are strictly dominant — better *and* roughly 1.7–4× cheaper than frontier solo (0.59×/0.24× the cost per solution). For Qwen and Kimi, orchestration buys the largest quality gains in the study at 2–3.5×

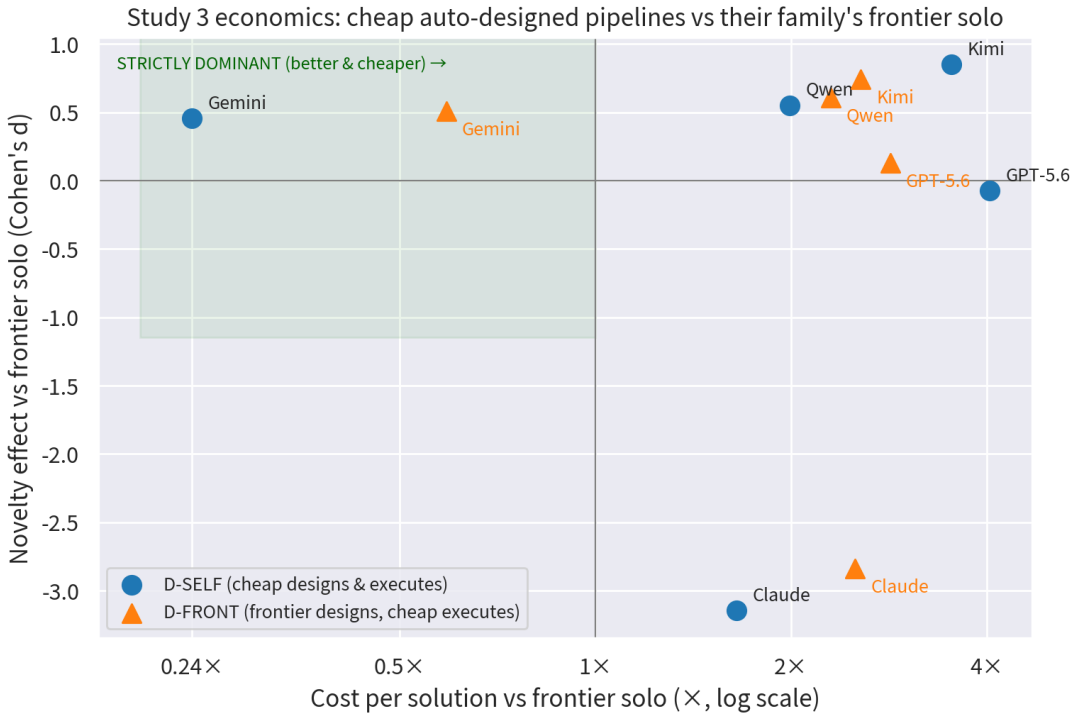



Figure 15: Study 3 economics: cost per solution versus Novelty effect, both relative to the family's frontier solo (F-S-N). The shaded quadrant (better *and* cheaper) contains Gemini on both arms — strict dominance. Qwen and Kimi buy quality at 2–3.5× cost; GPT-5.6 ties on quality at higher cost; Claude loses on quality at higher cost (Table 7).

solo cost (which remains low in absolute terms). For Claude, no cheap-executed pipeline approaches the flagship: buy solo. For GPT-5.6, quality is a tie and solo is cheaper. And because authorship effects are negligible (Contrast A), when orchestration is indicated the cheap model can design for itself — paying a frontier model to author the pipeline adds no measurable quality, except where the cheap designer itself is fragile (21 of 57 Kimi self-design attempts — 37% — failed at the design stage, making frontier- or human-authored templates the safer default for that family). Section 4.3.1 stress-tests this table on new problems: the extremes (Kimi, Claude) transport, the mid-tier rows do not, so the table should be read as an illustration of the *method* rather than a durable lookup.

> **Key Takeaway:** who *designs* the pipeline does not matter — models author better pipelines than our human instrument, and cheap models author as well as frontier ones. Whether to *use* a pipeline depends on the family: orchestration is a substitute for tier where the frontier ceiling is low (Gemini, Qwen, Kimi) and a dominated cost where it is high (Claude).

### 4.3.1 Out-of-sample confirmation: does family moderation generalize? (Protocol 16)

**Design.** Study 3 estimated family-specific orchestration effects on the 19 confirmatory problems. Protocol 16 (pre-registered OSF zkm8v, registered and timestamped before any generation) tests whether those estimates transport to *new* problems. We ran the D-SELF arm only — pre-declared, since Study 3 Contrast A found authorship effects negligible in 4/5 families — on 30 fresh wave problems × 3 replications × the same five

core families = 450 planned runs; 441 completed with 9 disclosed attrition events (Gemini 90/90, Claude 89/90, Qwen 89/90, GPT-5.6 88/90, Kimi 85/90; feasibility gate G1 passed for every family). The comparator is the *frozen* wave frontier-solo (F-S-N) baseline — zero new frontier-solo generation, so the test is a pure transport test of Study 3's effect sizes. Solutions were judged by the same frozen three-judge panel under the three verbatim prompt variants (3,963/3,969 valid unique judgments; 6 judge calls returned persistently unparseable content on three Claude solutions and are disclosed as 0.15% judged-missing attrition). The analysis script was frozen and hash-locked (SHA-256 recorded in the lab manifest) *before* judging began. Five binding gates translate the Study 3 point estimates into directional predictions: R1 Kimi $d > 0$; R2 Qwen $d > 0$; R3 Gemini $d > 0$; R4 GPT-5.6 equivalence (TOST, 90% CI on $d$ within $\pm 0.30$); R5 Claude $d < -1.0$. The headline claim "family moderation replicates" requires R1 $\wedge$ R2 $\wedge$ R3 $\wedge$ R5.

**Results: the extremes replicate; the middle does not.** Table 8 reports the gate outcomes. Two of five gates pass, and they are the two *extreme* families: Kimi's cheap auto-designed pipelines again beat frontier solo, by an even larger margin than predicted ($d = +1.25$ vs. $+0.85$ predicted, one-sided $p < 10^{-4}$), and Claude's again lose badly ($d = -1.90$ vs. $-3.15$ predicted, well past the $-1.0$ threshold). The three middle families fail: Qwen's predicted gain vanishes ($d = -0.23$), and Gemini and GPT-5.6 *reverse sign* — their self-designed pipelines score significantly *below* frozen frontier solo on the new problems (Gemini $d = -0.83$; GPT-5.6 $d = -0.88$, 90% CI $[-1.32, -0.43]$, so the R4 equivalence gate fails in the negative direction, not toward zero). The headline conjunction is therefore **not replicated** (2/5 gates).

Table 8: Study 3 confirmatory replication (Protocol 16): binding gate outcomes. $d$ = Cohen's $d$ on Novelty, D-SELF vs. frozen wave F-S-N, problem-level Welch tests, $n = 30$ problem pairs per family. Predictions are the Study 3 point estimates.

| Gate | Family | Predicted | Observed $d$ | 90% CI | Verdict |
|---|---|---|---|---|---|
| R1 | Kimi | $d > 0$ ($+0.85$) | $+1.25$ | $[+0.79, +1.72]$ | **PASS** |
| R2 | Qwen | $d > 0$ ($+0.55$) | $-0.23$ | $[-0.66, +0.19]$ | fail |
| R3 | Gemini | $d > 0$ ($+0.46$) | $-0.83$ | $[-1.28, -0.39]$ | fail (reversed) |
| R4 | GPT-5.6 | $\lvert d \rvert < 0.30$ | $-0.88$ | $[-1.32, -0.43]$ | fail (negative) |
| R5 | Claude | $d < -1.0$ ($-3.15$) | $-1.90$ | $[-2.41, -1.38]$ | **PASS** |

**Interpretation.** Family moderation is real but not a stable lookup table. The qualitative ordering — orchestration substitutes for tier where the frontier ceiling is low and is a dominated cost where it is high — survives at its extremes (Kimi, Claude), where effects are large. But for mid-tier families the sign and magnitude of the orchestration effect are problem-set-dependent: point estimates from 19 problems did not transport to 30 new problems drawn from the same generator. Combined with fast model turnover (the underlying model versions themselves change within months), any static per-family routing table is obsolete on arrival. The correct deliverable is therefore not a table but a *method*: cheap-tier benchmarking-as-routing — test the user's actual model on the user's actual problem type at trivial cost (a D-SELF probe run costs \$0.01–\$0.73 per solution; Table 7) and route on the result.

**Why we did not extend to new families.** A natural next step — repeating the study on the five additional families used in the wave replication — was considered and deliberately not registered. The scientific question it would answer (does family mod-

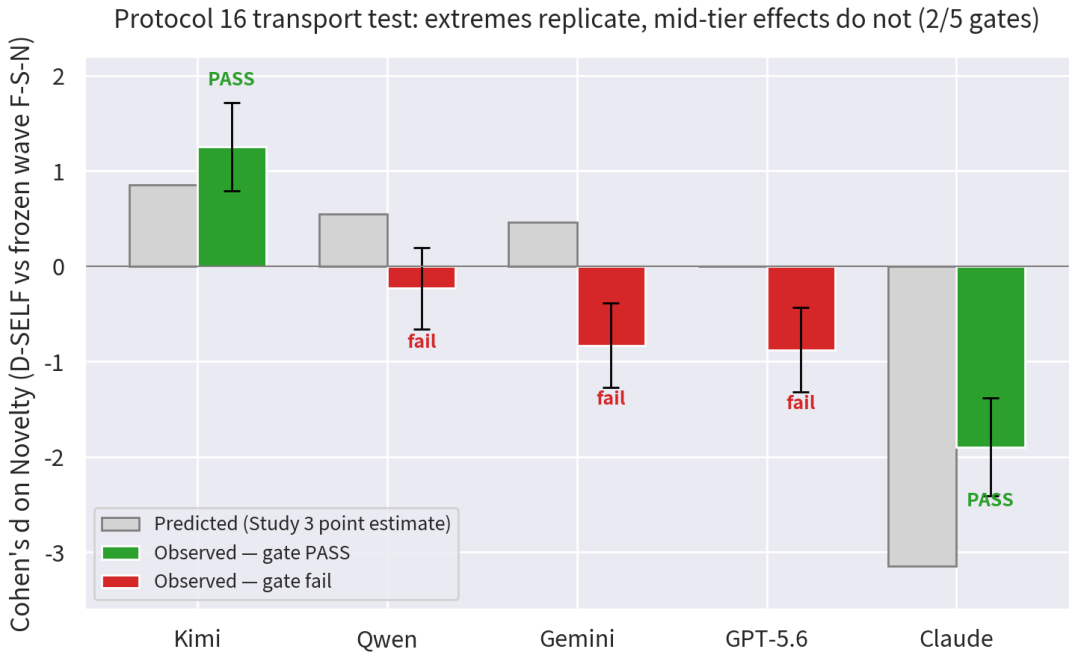



Figure 16: Protocol 16 transport test: predicted (Study-3 point estimates) versus observed Cohen's $d$ on Novelty against the frozen wave F-S-N baseline, with 90% CIs. The extreme families pass their directional gates (R1 Kimi, R5 Claude); the three mid-tier families fail (Table 8).

eration exist for new families?) is already answered affirmatively by Studies 1–3; and Protocol 16's central finding is that mid-tier effect sizes do not transport even *within* known families across problem sets. Five additional families would therefore add point estimates subject to the same non-transport, at roughly 450 runs and 4,000 judgments of cost, without changing the methodological conclusion. We flag a bounded generalization probe on new families as future work, positioned as a product-method validation (benchmarking-as-routing) rather than a confirmatory experiment.

**Key Takeaway:** the Study 3 extremes replicate on new problems — cheap pipelines beat frontier solo where the frontier ceiling is low (Kimi, $d = +1.25$) and lose where it is high (Claude, $d = -1.90$) — but mid-tier effects do not transport across problem sets. Orchestration routing must be measured per deployment, not looked up: benchmark-as-routing, not a routing table.

**Strategic Implication:** Do not copy this paper's routing table into your budget — build your own. Only the extremes survived new data: if your stack sits at the low-ceiling end (Kimi-class), cheap auto-designed pipelines beat the flagship outright — default to orchestration; if it sits at the high-ceiling end (Claude-class), pay for the flagship solo and skip pipelines entirely. For everything in between, the effect flips sign across problem sets, so route by measurement, not by lookup: run a cheap self-designed probe on your own tasks ($0.01–$0.73 per solution; Table 7) and let the result decide. Routing is an ongoing measurement practice, not a one-time purchase decision.

Table 9: Summary of key findings across the three studies and their pre-registered replications.

| Finding | Evidence | Section |
|---|---|---|
| Team structure helps lower-tier models more than frontier ($H_1$; persona-conditional, $\approx 0$ in neutral cells) | $d = -0.227$ | Section 4.1.1 |
| Frontier > cheaper overall ($H_2$) | $d = +0.294$ | Section 4.1.1 |
| Teams slightly beat solos on average ($H_3$) | $d = +0.117$ | Section 4.1.1 |
| Persona penalty on net ($H_4$, reversed) | $d = -0.691$ | Section 4.1.1 |
| Structure × persona interaction ($H_5$, n.s.) | $F = 1.23$, $p = 0.27$ | Section 4.1.1 |
| Persona-loaded cells Pareto-dominated | cost–quality plane | Section 4.1.2 |
| CTP < FSN in 3 of 5 families (large); borderline in kimi; null in gpt | paired comparisons | Section 4.1.3 |
| Refinement recovers 64% of gap where a real gap exists (qwen); G2 rejected | Protocol 09 tests (H2.1–H2.3) | Section 4.2.1 |
| Orchestration buys coverage; frontier wins on density | $\kappa = 0.824$; 13 codes | Section 4.2.2 |
| Pareto frontier stable across $\varphi$ | {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N} | Section 4.1.4 |
| *Replicated (wave):* team effect, upgrade threshold, bimodal economics | W1/W2/W6 PASS, 10 families, $n = 2{,}418$ | Section 4.1.7 |
| *Exploratory (gate FAILED in wave):* domain contingency; capability-gap mechanism; marketing variance advantage | W3/W4/W5 FAIL | Section 4.1.6, Section 4.1.7 |
| *Exploratory (not wave-tested; wave excluded persona cells):* persona novelty gain is solo-only; teams squash it | D45 battery | Section 4.1.6 |
| Design authorship negligible (4/5 families); auto-designed > human-designed pipelines (5/5) | Study 3, Protocol 13 | Section 4.3 |
| Cheap auto-designed pipelines beat frontier solo for Gemini/Qwen/Kimi; lose for Claude; tie GPT | $d$ up to +0.85; cost 0.24–3.5× solo | Section 4.3 |
| *Partially replicated:* extremes transport to new problems (Kimi $d = +1.25$; Claude $d = -1.90$); mid-tier families do not — routing must be measured, not tabulated | Protocol 16, 2/5 gates, $n = 441$ | Section 4.3.1 |

## 4.4 Summary of Key Findings

# 5 Discussion

## 5.1 The Density Ceiling: Why Additive Pipelines Hit a Wall

The qualitative mechanism (Section 4.2.2) explains the pattern. Orchestration is compositional — add a critic, add a refiner, add a section, add a table. Composition grows the answer but cannot compress it. The frontier's residual advantage sits exactly where additive composition fails: `concise_maxims` (compressed, quotable strategic principles) and zero `redundancy_bloat`. These are the outputs of a single model that has been selected or trained to produce dense prose. An add-only refiner, by design, cannot delete the boilerplate the team structure produced. The falsifiable prediction that follows is that a *subtractive* refiner — one permitted to cut as well as add — would narrow the density gap. This is the central future-work experiment implied by our results.

## 5.2 A Decision Framework for AI Investment

The results support a contingency framework for AI investment rather than a universal rule, and the pre-registered replications discipline every rung of it: we recommend only what survived contact with new data. For budget-constrained firms, **invest in the lower tier first**: a single call from a cheaper model (C-S-N) achieves 98% of frontier quality at 23% of the cost (T-EGR 6.15) — the highest-return configuration in the study, replicated out-of-sample across 10 families (Section 4.1.7). **Reserve frontier solo for density-critical output** (executive briefs, compressed strategic principles): the frontier advantage is real, replicated (W2), and concentrated exactly where additive composition fails. **Treat pipelines as a measured, family-conditional option — not a default**: the fixed three-step pipeline tested in Study 1 (C-T-N: 98.8% of frontier quality at 88% of the cost, T-EGR 1.49) did *not* replicate as an efficient cell — on fresh problems it is the dominated middle (T-EGR 0.96, excluded from the Pareto set; Section 4.1.7) — and auto-designed pipelines beat that fixed human design in all five families (Section 4.3). Where a cheap auto-designed pipeline beats its own family's frontier solo (the low-ceiling stacks Gemini, Qwen, Kimi), it becomes the quality-maximizing choice; where the ceiling is high (Claude), it is a dominated expense; and mid-tier effects do not transport across problem sets, so the rung is operated by measurement, not by table (Section 4.3.1). **Use a low-cost refinement pass as a risk-mitigation strategy** when absolute quality parity is the target: the refined kimi point reaches the pooled F-S-N mean (4.585 vs. 4.586) and exceeds the weakest frontier family (qwen, 4.557) at 78% of F-S-N's cost, the lowest-cost route to that milestone among the configurations tested — noting that the paired within-family comparison remains negative (Section 4.2.1). **Avoid personas**: they degrade quality sharply while reducing cost only marginally — a value-destroying configuration. **Do not orchestrate at the frontier tier**: frontier team cells spend 3.5× the frontier-solo cost for +0.024 points of quality (T-EGR 0.27), making them the worst dollar in the experiment.

The family-conditional detail behind the pipeline rule (Section 4.3, Table 7): where the deployment stack is Gemini, the auto-designed cheap pipeline is strictly dominant over frontier solo (better novelty at 0.24× the cost); for Qwen and Kimi the pipeline is the quality-maximizing choice at 2–3.5× solo cost; for Claude the solo flagship remains untouchable and pipelines are a dominated expense; and in all cases the design fee is better spent on execution than on authorship — self-designed pipelines match frontier-designed ones in four of five families.

## 5.3 Contribution to the Business of AI Literature

Our findings extend the author's prior lineage in two directions. The EGR metric introduced in Hariri (2026b) measured the cost performance of *personas in single models*; T-EGR extends it to teams and pipelines, preserving the ratio form while re-basing on the frontier cell. The team-composition contingency identified on verifiable tasks in Hariri (2025b) reappears here on open-ended tasks, but with the sign reversed for personas (harmful rather than neutral on substance). The jagged-frontier finding of Dell'Acqua et al. (2023) is replicated as a *cell-level* result: lower-tier models inside the persona-free, coverage-oriented region match frontier quality within 1–2 points of the scale at a fraction of the cost; cheaper models pushed outside by persona-loading or by density demands fall below the frontier.

The test-time-compute prediction of Snell et al. (2024) is partially confirmed: more

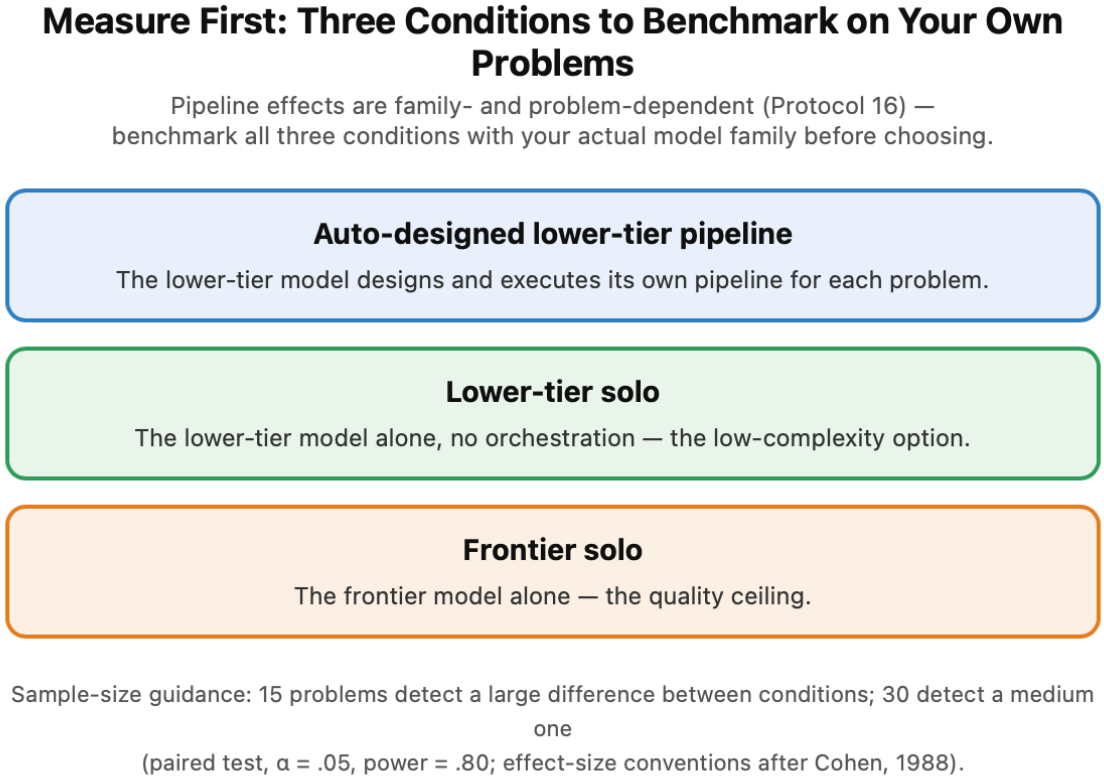



Figure 17: Practitioner contingency rules. Decision flowchart mapping budget tier and task type to the recommended orchestration cell, with expected T-EGR. Derived from the Section 5.2 contingency model.

inference compute (team, refinement) improves the cheaper model, but only up to a parity ceiling, and only when priced in dollars does the tradeoff reveal its practitioner shape. Our contribution is the dollarization of the compute-optimal curve on open-ended tasks where no verifier exists.

### 5.4 Methodological Lessons for AI Deployment Research

Two lessons carry beyond this study. First, preregistered negative superiority results can be the spine of a paper, not its appendix. The G2 mechanical failure in Section 4.2.1 is as informative as any positive finding, and preregistration is what made it reportable as a primary result rather than as a null. Second, token-invariant costing — storing tokens and re-pricing through a frozen vector — is reusable infrastructure for the field: every reported dollar is reproducible by any reader who supplies an alternative price vector (Diewert, 1976; Sandve et al., 2013).

## 6 Managerial Implications

This section translates the empirical findings into decisions a budget owner can act on. Where Section 5.2 states the contingency framework as a research result, the guidance below is written as practice: a sequenced budget rule, an operating model, a risk register,

and a sourcing decision. Every claim carries its dollar figure first and its supporting statistic second, and points to the quick-reference table (Table 10) rather than repeating it.

## 6.1 The Investment Ladder: A Sequenced Budget Rule

The central managerial finding is that AI capability should be *purchased in a sequence*, not all at once. Spend down the following ladder and stop at the rung where the next dollar buys less quality than it costs:

1. **Default to a single lower-tier call (C-S-N).** It delivers 98% of frontier quality for 23% of the price (T-EGR 6.15) — the highest-return dollar in the study and the correct default for most open-ended knowledge work.
2. **Add a pipeline only where measurement supports it (family-conditional).** The fixed three-step team tested in Study 1 reached 98.8% of frontier quality at 88% of the cost (T-EGR 1.49), but that cell did not replicate as efficient — on fresh problems it is the dominated middle (T-EGR 0.96; Section 4.1.7) — and auto-designed pipelines beat the fixed design in all five families (Section 4.3). Where a cheap auto-designed pipeline beats its own family's frontier solo (low-ceiling stacks), it is the quality-maximizing rung; elsewhere skip it. The marginal dollar now buys *coverage*, not density.
3. **Buy a refinement pass as cheap insurance ($0.001–0.003, family-dependent).** It lifts a refined lower-tier answer to the weakest frontier family's level (qwen, 4.557) at 78% of frontier-solo cost — but it never achieves superiority. Treat it as a risk-mitigation rung, not a replacement rung.
4. **Reserve frontier solo (F-S-N) for density-critical output** — executive briefs and compressed strategic principles, where the residual frontier advantage actually lives (Section 4.2.2).
5. **Never orchestrate at the frontier tier (F-T-N).** A frontier team costs 3.5× the frontier-solo price (353% of cost) for negligible quality gain (T-EGR 0.27) — the worst dollar in the experiment.

**Strategic implication.** The budget question is not "frontier or cheap?" but "which rung, for which task?" A firm that routes coverage work to the lower tier and reserves the frontier for density captures most of the quality at a fraction of the cost; a firm that pays frontier prices for team orchestration overpays for output it could have bought cheaper.

## 6.2 An Operating Model: Route by Measurement, Not by Table

Study 3 changes how the ladder is operated. The economics of orchestration are *family-specific* (Section 4.3, Table 7): the auto-designed cheap pipeline strictly dominates frontier solo on a Gemini stack at 0.24× the cost, maximizes quality for Qwen and Kimi at 2–3.5× solo cost, and is a dominated expense on Claude, where the solo flagship is untouchable. The managerial consequence is direct: **there is no universal routing table a firm can copy.** The operating model is a *benchmark-and-route* discipline — on a small held-out set of the firm's own tasks, measure the cheap-pipeline-versus-frontier-solo T-EGR for the specific model family in the stack, then route on the measured number.

Two further operating rules follow. First, **spend the design fee on execution, not authorship**: self-designed pipelines matched frontier-designed ones in four of five families, so a firm does not need to pay a frontier model to author its workflows. Second,

**treat routing as a recurring procurement KPI**, re-measured whenever a model version or an API price changes — because the dollar figures that drive the ladder move with the market, even when the quality ordering does not. Third, **do not hand-design pipelines**: model-authored pipelines beat our human-designed three-step instrument in all five families, and the auto-designer supplies richer structure than a fixed template (critic-gated feedback loops in 98.9–100% of designs, five to six steps on average). The one caution is design-stage fragility: 37% of Kimi self-design attempts failed schema validation, so for fragile designers keep validated frontier- or human-authored fallback templates.

## 6.3 Risk Register: What Can Go Wrong and How to Bound It

- **Persona risk.** Role/persona prompting degrades substance while reducing cost only marginally (Section 4.1.2) — a value-destroying configuration that is easy to adopt by accident because it "feels" more professional. *Control:* audit persona usage and strip it wherever substance, not style, is graded.
- **Density risk.** Orchestration buys coverage but not density, and a refinement pass closes most of the gap where a real gap exists (qwen, 64%) without ever beating the frontier (Section 4.2.2, Section 4.2.1). *Control:* do not expect a cheap additive pipeline to substitute for compressed, quotable insight; buy the frontier for that class of output.
- **Replication risk — honestly bounded.** The core economics replicated out-of-sample across 10 model families (Protocol 12). For Study 3's family rules, the *extremes* transported to new problems (Kimi pipelines again beat frontier solo; Claude's again lost badly), but the *mid-tier* effects did not (Protocol 16 passed 2 of 5 pre-registered gates). *Control:* treat the headline rules as bankable and the family-specific rules as hypotheses to re-measure locally, not as guarantees. This is the payoff of pre-registration — we can tell the manager which findings survived a fresh bet and which did not.

## 6.4 Build-versus-Buy: A Capability-Sourcing Decision

Frontier models are *bought*: a fixed per-call cost for a given quality level. Orchestration pipelines are *built*: an upfront architectural investment that lowers the marginal cost per call. The build path pays off when (a) the task is coverage-sensitive, (b) the quality bar is "good enough" rather than "best available," and (c) the firm can absorb the engineering cost of maintaining a multi-step pipeline — a condition that auto-design largely dissolves: Study 3 shows models author better pipelines than our hand-designed instrument, at a design fee that is a small fraction of the first execution (2–3% of total tokens), so the upfront architectural investment collapses to a single design call. When density is the binding constraint, the buy path (frontier solo) remains optimal.

The decision statistic is T-EGR: **build when the measured T-EGR on your own workload exceeds 1, buy when the frontier's density premium is the thing you actually need.** Critically, this is a *per-task-family* decision, not a firm-wide posture — the same organization should build cheap pipelines for coverage work and buy frontier calls for density work, and should expect the mix to shift as model prices and capabilities move.

# 7 Limitations

Four limitations bound the claims; study-specific disclosures (the blended-$\varphi$ price assumption with its sensitivity panel, list-price costing and unpriced latency, wave execution events, Study-3 attrition and retry cost accounting, and the unreplicated persona-carried $H_1$ interaction) are stated where they arise in Methodology and Results.

**(1) LLM-only evaluation.** No human panel was recruited. Our bias controls (blinding, order shuffle, disjoint judge families, three prompt variants) follow the best current practice (Zheng et al., 2023; Verga et al., 2024) but do not substitute for human agreement on a subset.

**(2) One task domain.** All stimuli are English business-strategy problems; generalization to code, scientific writing, or non-English domains is an empirical question.

**(3) Study 2 rests on a single sample.** Unlike Studies 1 and 3, whose headline claims were stress-tested on fresh problems in pre-registered replications (Section 4.1.7, Section 4.3.1), the refinement ceiling and the coverage-versus-density mechanism were estimated on the original locked sample only (*cf.* the replication-scope note in Section 4.2.2). A confirmatory refinement wave is future work.

**(4) Study-3 domain breakdowns are descriptive.** The family × domain patterns reported in Section 4.3 rest on 2–5 problems per cell, underpowered for inference by design; they are indicative patterns, not tested findings. Powering domain-level claims was not the purpose of Study 3.

# 8 Future Directions

Two priorities named in earlier drafts are now complete: the out-of-sample confirmatory replication wave (Section 4.1.7) — which promoted the core claims (team effect W1, upgrade threshold W2, economics W6) and rejected the exploratory upgrades — and pipeline design authorship (Study 3, Section 4.3), which executed the registered A7 arm and found authorship effects negligible and orchestration economics family-specific. Two candidates from the original roadmap are closed by the results themselves: **loop control** (Protocol 14) is superseded because Study 3 found critic-gated loops in 98.9–100% of auto-designed pipelines yet outcomes remained family-determined, and **static routing** (Protocol 15) is superseded because Protocol 16 showed mid-tier effects do not transport across problem sets. Four larger research programs follow.

**(1) The editing-operations program.** The density ceiling (Section 4.2.2) says the binding constraint is *what operations* a pipeline may perform, not how many steps it has: our add-only refiner could grow answers but never compress them. The open program is to study editing operations directly — a refiner permitted to cut, compress, merge, and reorder — and to test whether models can learn to allocate compute between generation and deletion. This is the direct falsifiable test of the density mechanism and the natural successor to Study 2.

**(2) Frontier-team outputs as an object of study.** The worst dollar in the experiment (frontier teams: 3.5× the cost for negligible gain) is unexplained. We hold the frozen frontier-team outputs from Study 1; nobody has characterized *what* they actually produce — whether teaming dilutes the frontier's density the way it dilutes persona novelty, or whether the outputs are genuinely richer but not better. A descriptive and qualitative analysis of these existing outputs would explain the anomaly and inform whether frontier orchestration deserves further study at all.

**(3) Benchmarking-as-routing: validating the measurement method.** Protocol 16 killed static routing tables; the open project is the *method* itself: does a cheap self-designed probe on a user's own tasks reliably predict the correct routing decision? This is positioned as a product-method validation (the research-to-product bridge for the DeeWaanAI platform), not a confirmatory experiment.

**(4) Semantic claim-to-evidence verification.** Building this paper's machine-readable claim map taught us a methodological lesson: value fidelity (every number traces to frozen data) is necessary but not sufficient — numbers can match their sources while the prose misstates what they mean (two such episodes were caught and corrected in this project). Extending claim-to-evidence mapping from values to semantics — direction, attribution, and existence-of-analysis checks on interpretive prose — is a reusable methodology for AI-assisted research and a study of its own.

# 9 Conclusion

Can a team of cheaper models beat a frontier model working alone? Across three pre-registered studies and two out-of-sample replications (5,303 runs, 10 model families, 49 business problems), the answer is **contingent — on the family's tier price gap (3.1–8×) and on the model family itself.** Pooled, the cheaper team reaches 98.8% of frontier-solo quality at 88% of its cost — short of parity, with the residual gap concentrated in density, which additive pipelines cannot manufacture. Yet auto-designed cheaper pipelines beat frontier solo outright for three of the five families (kimi, qwen, gemini), at costs ranging from 0.24× to 3.5× the frontier-solo price; the remaining two favor frontier solo — claude on quality (the pipeline reaches 83%), gpt-5.6 on cost (quality ties at 4× the price). Because token usage is comparable across tiers, the cheaper solo's advantage — 98% of frontier quality at 23% of its cost — is the direct monetization of the tier price gap and the highest-return default in the study. And since pipeline effects are family- and problem-dependent, and model-authored designs outperform our human-designed instrument, configuration choices should rest on measurement against the firm's own tasks.

The paper makes three contributions. *Empirically*, it provides the first dollarized, pre-registered, cost-matched test of the substitution claim on open-ended knowledge work, and its replication gates publicly separated what survives new data from what does not — the core economics replicated; the contingency nuances did not. *Methodologically*, it contributes reusable rigor infrastructure: token-invariant costing through a frozen price vector (every dollar figure re-derivable under alternative prices) and a machine-readable claim-to-evidence map tracing every printed statistic to frozen raw data. *Practically*, it converts orchestration from an engineering project into a measurement practice: in all five families, models author better pipelines than our hand-designed instrument, and paying a frontier model to do the designing adds no measurable quality — what matters is measuring the pipeline effect on the firm's own model stack and tasks.

These findings are bounded by three limitations: the task domain is English business strategy; evaluation rests on an LLM judge panel without a human reference; and pipeline effects vary by model family and problem set. Future research should test editing operations beyond addition, characterize frontier-team outputs, and validate benchmarking-as-routing as a method on firms' own tasks.

Table 10: Managerial quick-reference: recommended configuration by task profile.

| **Task profile** | **Cell** | **Quality** | **Cost** | **T-EGR** | **Override when...** |
|---|---|---|---|---|---|
| Coverage-heavy (plans, timelines, matrices) | C-S-N | 98% | 23% | 6.15 | absolute quality non-negotiable → F-S-N |
| Density-critical (exec briefs, strategic principles) | F-S-N | 100% | 100% | 1.00 | budget tight → C-S-N at 98% |
| Pipeline option (auto-designed, family-conditional) | D-SELF | beats solo at extremes | 0.24–3.5× | — | mid-tier stack → measure first (Section 4.3.1) |
| Budget-minimum, quality-adequate | C-S-P | 92% | 17% | 8.60 | novelty needed, substance traded |
| Refinement as risk mitigation | Refined C-T-P | ∼100% | 78% | — | never achieves paired superiority |
| **Avoid** | F-T-N | 100.5% | 353% | 0.27 | worst dollar in the experiment |
| **Avoid** | F-T-P | 95.3% | 365% | 0.27 | persona penalty on top of team overhead |

# A Executive Summary for Practitioners

**Decision context.** A firm that needs LLM-generated knowledge work can either (a) pay for a frontier model or (b) use lower-tier models, optionally orchestrated into a multi-step pipeline. Which dollar buys more quality?

**Bottom line.** Across 1,900 runs, five model families, and 19 business problems: a single call from a lower-tier model delivers **98% of frontier quality at 23% of the cost** (T-EGR 6.15). Orchestrating that model into a three-step builder–critic–refiner pipeline adds coverage for about 3.8× that cost (T-EGR 1.49). Orchestrating a frontier model costs **3.5× more for negligible gain**. A fraction-of-a-cent refinement pass ($0.001–0.003, family-dependent) recovers most of the remaining gap where a real gap exists (qwen, 64%) but never beats the frontier.

**Three rules for AI budget allocation:**

1. **cheaper tier, single call** → highest ROI (T-EGR 6.15). Pipelines are a measured, family-conditional option, not a default: auto-designed cheap pipelines beat the frontier flagship on low-ceiling stacks (Gemini, Qwen, Kimi) and lose where the ceiling is high (Claude) — the fixed three-step pipeline tested here did not replicate as an efficient cell, so measure on your own tasks before committing.
2. **Frontier tier (solo)** → best absolute quality per dollar when density matters (compressed insight, zero boilerplate).
3. **Never add personas as a quality lever** — they degrade judged quality sharply ($d = -0.691$) while reducing cost only marginally.

**Caveat.** Cheaper configurations buy *coverage* (more structure, more length); frontier models win on *density* (insight per paragraph). If the task rewards concise strategic principles, the frontier is worth the premium. If the task rewards completeness, the lower-tier configuration is the better bet.

**From findings to product.** This research was conducted on, and now feeds back into, the DeeWaanAI orchestration platform. Three findings have already changed the product: (1) the persona penalty ($d = -0.691$) removed personas as a default quality lever; (2) Study 3 established that pipeline *design authorship* is a negligible factor, so the platform's auto-designer lets the cheap model design for itself rather than billing a frontier design pass; and (3) Protocol 16's partial replication converted the static per-family routing table into *benchmarking-as-routing* — a cheap per-deployment probe (test the user's model on the user's problem type, then route), because mid-tier effects do not transport across problem sets. The loop is deliberate: the platform supplies the research instrument, the pre-registered findings supply the product's routing logic, and the disclosure in Section 3.9 keeps the relationship transparent.

# B Protocol and Arm Registry

A **protocol** is a pre-registered experiment plan frozen before its data are generated; an **arm** is one experimental condition within a protocol (Section 3). Table 11 lists every numbered protocol in the research program with its role and status; Table 12 defines every arm label used in the text. Protocols 05–08 were designed as *modifications of the original registration* rather than new studies: 05 corrected pilot issues before the main run; 06 extended the interim-2 analysis with a pre-registered robustness battery (items 6.1–6.7); 07 added post-lock exploratory statistics (items 7.1–7.5); 08 replaced the deductive codebook of Protocol 02 with an inductive one. In the decision log and analysis artifacts these four appear as "Amendments A1–A4"; the paper uses protocol numbers throughout.

Table 11: Protocol registry. "Frozen" = time-stamped before any data for that question were generated. OSF GUIDs are mapped in Table 2.

| Protocol | Role | Status |
|---|---|---|
| 01 | Study 1 factorial pre-registration ($H_1$–$H_5$) | frozen; executed |
| 02 | Qualitative analysis (deductive codebook) | frozen; superseded by Protocol 08 |
| 03 | Sequential analysis (O'Brien–Fleming interims) | frozen; executed |
| 04 | Cost-effectiveness analysis (T-EGR, Pareto) | frozen; executed |
| 05 | Modification of Protocol 01: pilot corrections | frozen; executed |
| 06 | Modification of Protocol 01: extended interim-2 robustness battery (items 6.1–6.7) | frozen; executed |
| 07 | Modification of Protocol 01: exploratory statistics (items 7.1–7.5) | frozen; executed |
| 08 | Modification of Protocol 02: inductive qualitative coding | frozen; executed |
| 09 | Study 2: refinement probe | frozen; executed |
| 10 | Study 2: qualitative coding | frozen; executed |
| 11 | Study 2 extension: refinement (GPT, Gemini; cross-family refiner) | frozen; executed |
| 12 | Confirmatory replication wave (30 fresh problems, 10 families) | OSF jg49x; executed |
| 13 | Study 3: pipeline design authorship (D-SELF vs. D-FRONT) | OSF bwc93/639eq; executed |
| 14 | Loop control (critic-gated cycles) | drafted; superseded by Study 3 (D59) |
| 15 | Static routing from problem text | drafted; superseded by Protocol 16 (D59) |
| 16 | Study 3 confirmatory replication (family-moderation transport) | OSF zkm8v; executed |

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Table 12: Arm registry. The eight factorial cells are tier (C = cheaper, F = frontier) × structure (S = solo, T = team) × persona (N = none, P = persona). A5–A8 are mechanism arms registered in Study 1; D-SELF/D-FRONT are the Study 3 arms.

| **Arm** | **Definition** | **Status** |
|---|---|---|
| C-S-N | cheaper solo, no persona | executed (Study 1 + wave) |
| C-S-P | cheaper solo, persona | executed (Study 1) |
| C-T-N | cheaper team (builder → critic → refiner), no persona | executed (Study 1 + wave) |
| C-T-P | cheaper team, persona | executed (Study 1) |
| F-S-N | frontier solo, no persona (baseline) | executed (Study 1 + wave) |
| F-S-P | frontier solo, persona | executed (Study 1) |
| F-T-N | frontier team, no persona | executed (Study 1) |
| F-T-P | frontier team, persona | executed (Study 1) |
| A5 | critic-gated WHILE loop on the cheap team | registered; never executed in Study 1 |
| A6 | cost-matched frontier-solo baseline | registered; never executed in Study 1 |
| A7 | adaptive per-problem pipeline designer | registered; executed as Study 3 |
| A8 | shuffled persona-to-role mapping | registered; never executed in Study 1 |
| D-SELF | cheap sibling designs *and* executes the pipeline | executed (Study 3 + Protocol 16) |
| D-FRONT | frontier flagship designs; cheap sibling executes | executed (Study 3) |


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